

Recurrent Neural Networks

Plan of Attack

Plan of Attack

What we will learn in this section:

- The idea behind Recurrent Neural Networks
- The Vanishing Gradient Problem
- Long Short-Term Memory (LSTM)
- Practical Intuition

- EXTRA: LSTM Variations

Recurrent Neural Networks

Supervised	Artificial Neural Networks	Used for Regression & Classification
	Convolutional Neural Networks	Used for Computer Vision
	Recurrent Neural Networks	Used for Time Series Analysis
Unsupervised	Self-Organizing Maps	Used for Feature Detection
	Deep Boltzmann Machines	Used for Recommendation Systems
	AutoEncoders	Used for Recommendation Systems

Recurrent Neural Networks

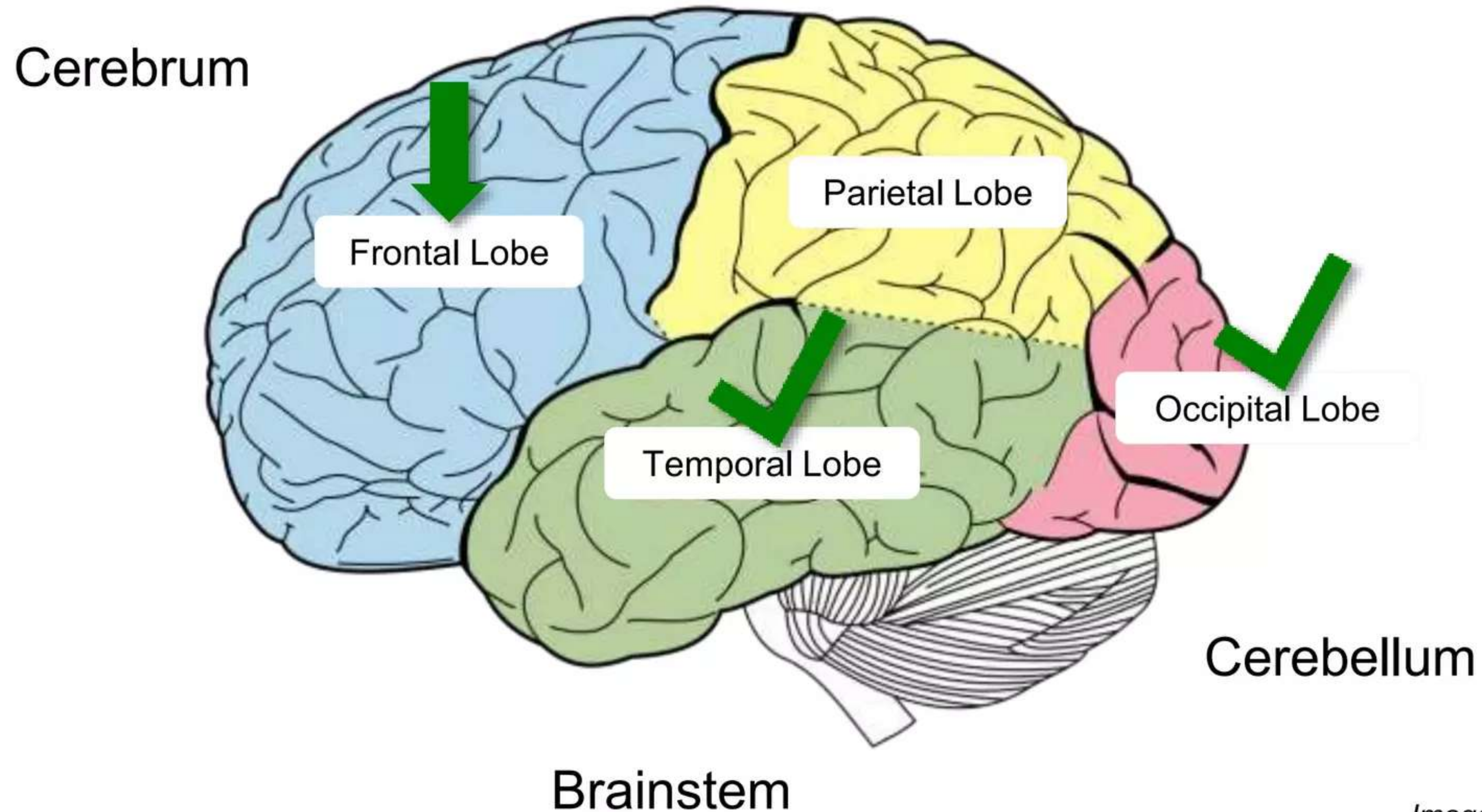
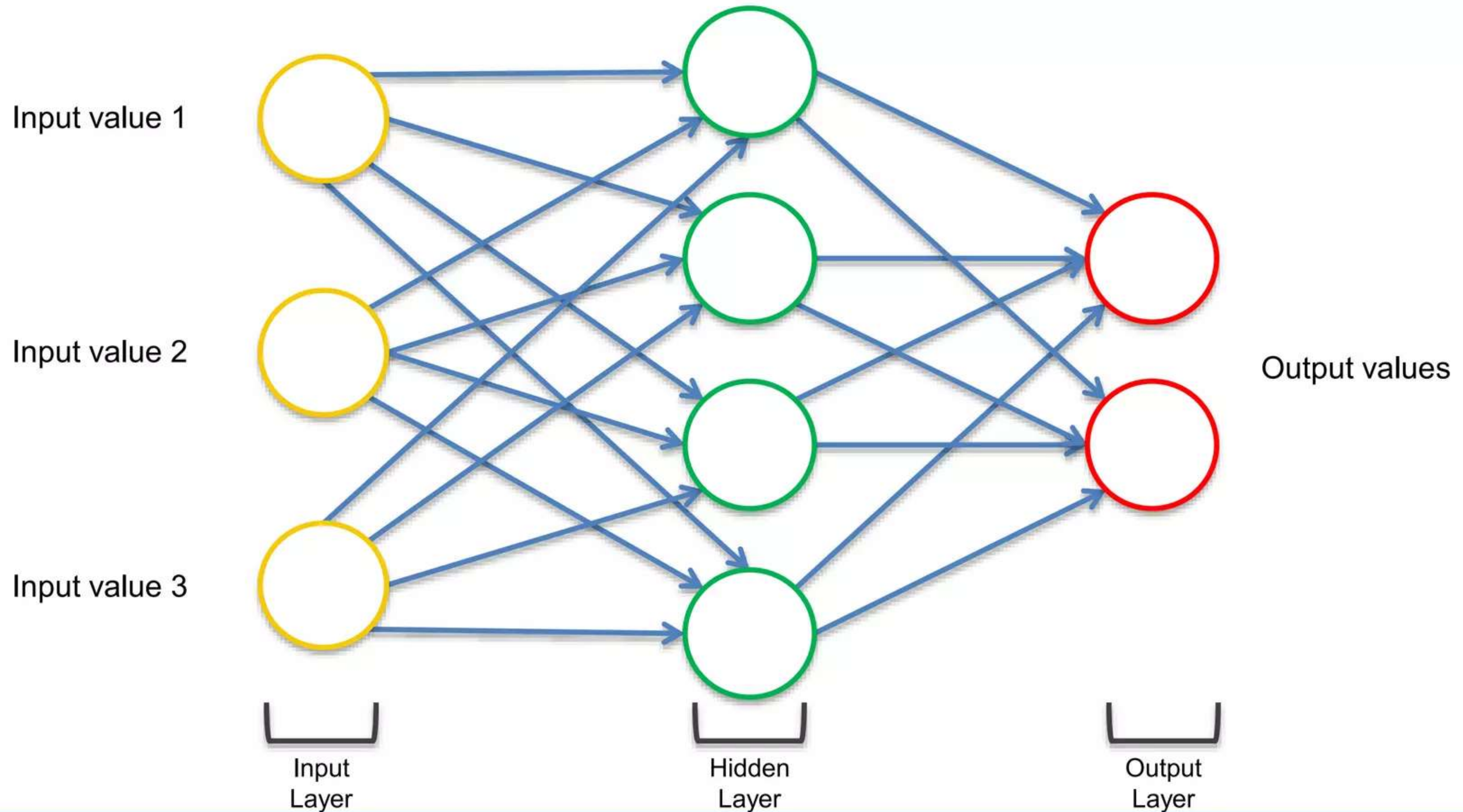


Image Source: Wikipedia

Recurrent Neural Networks

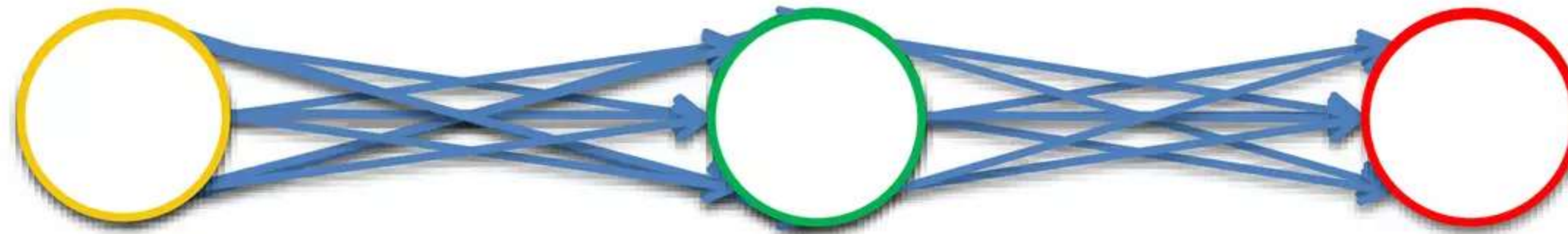


Recurrent Neural Networks

Input value 1

Input value 2

Input value 3



Output values



Input
Layer

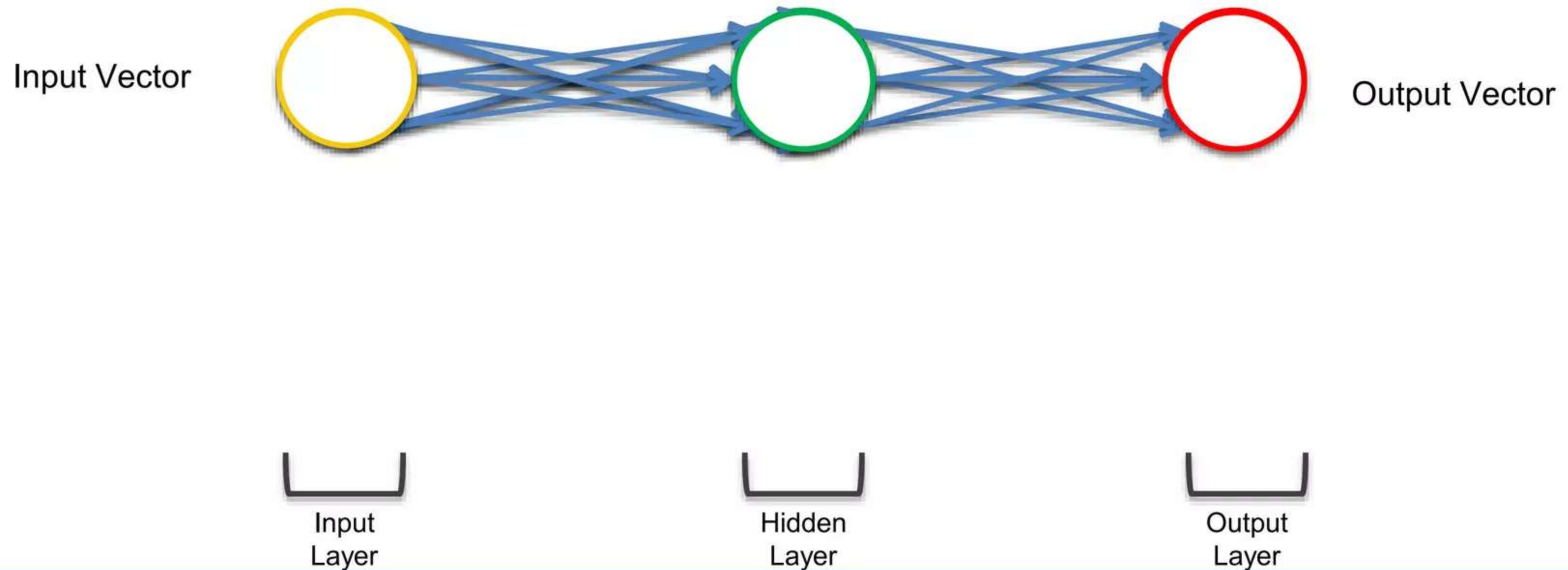


Hidden
Layer

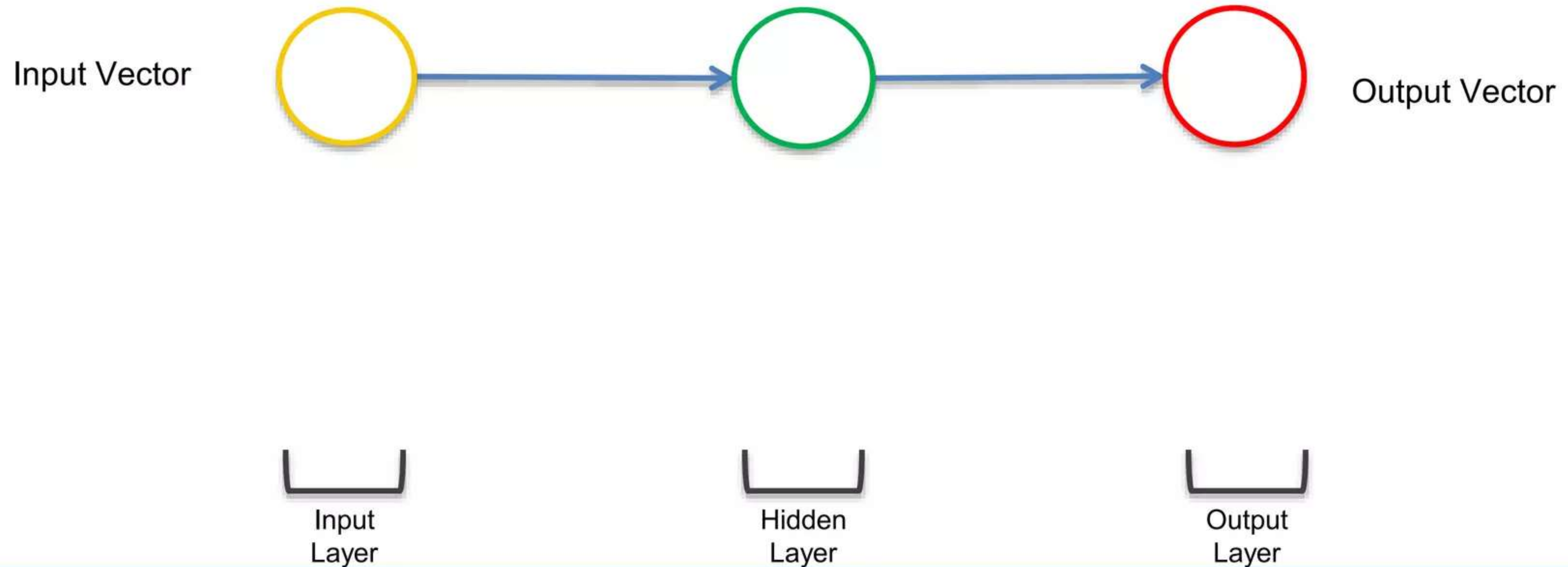


Output
Layer

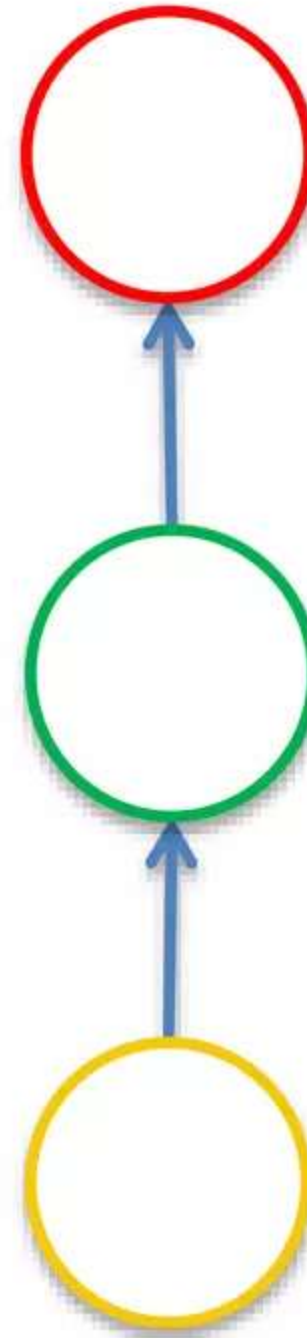
Recurrent Neural Networks



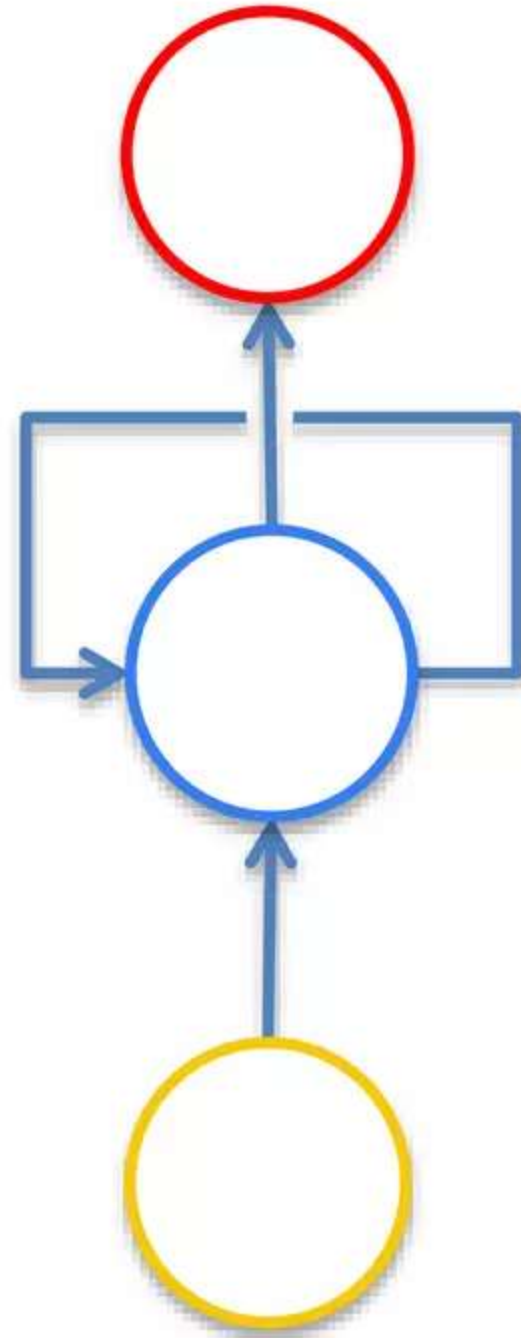
Recurrent Neural Networks



Recurrent Neural Networks



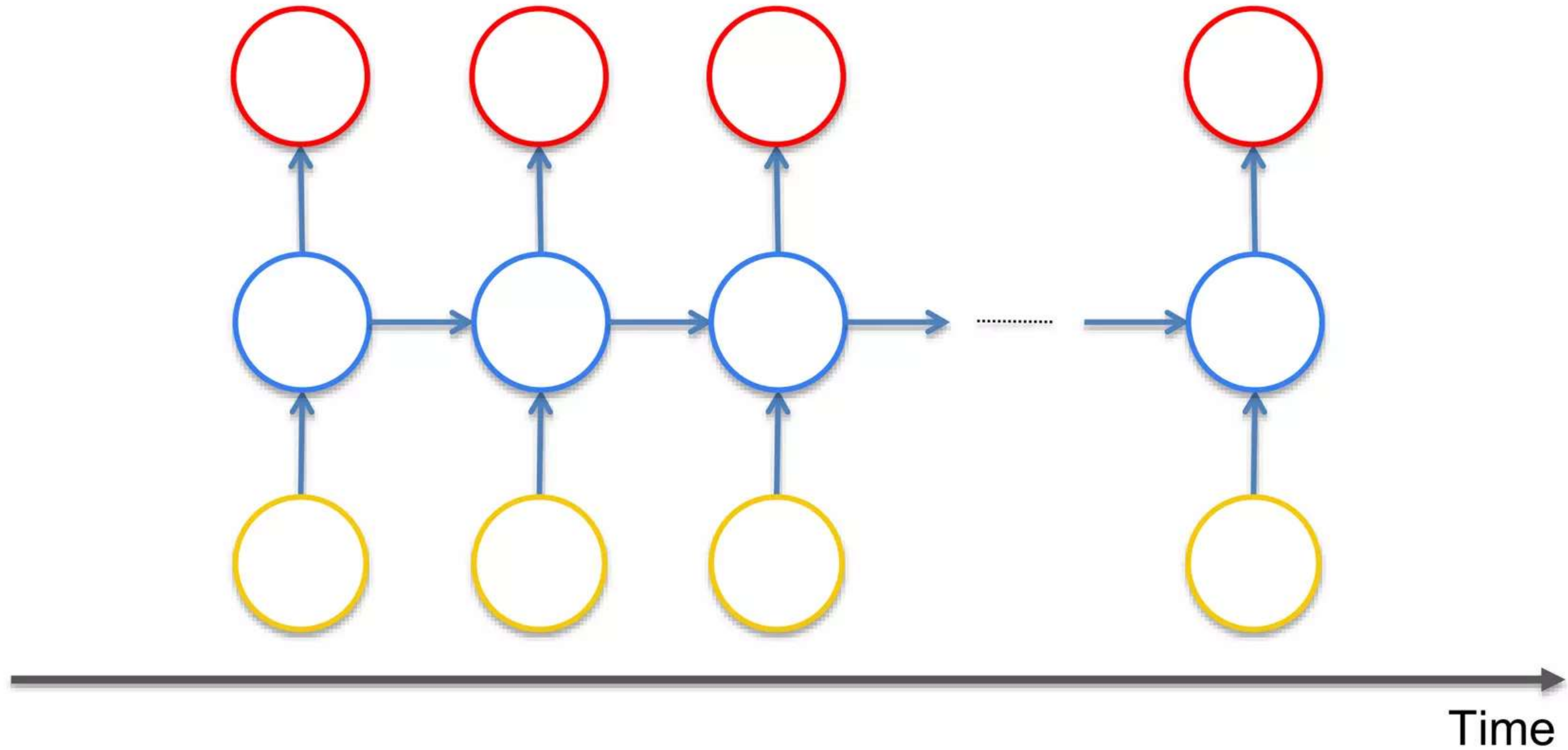
Recurrent Neural Networks



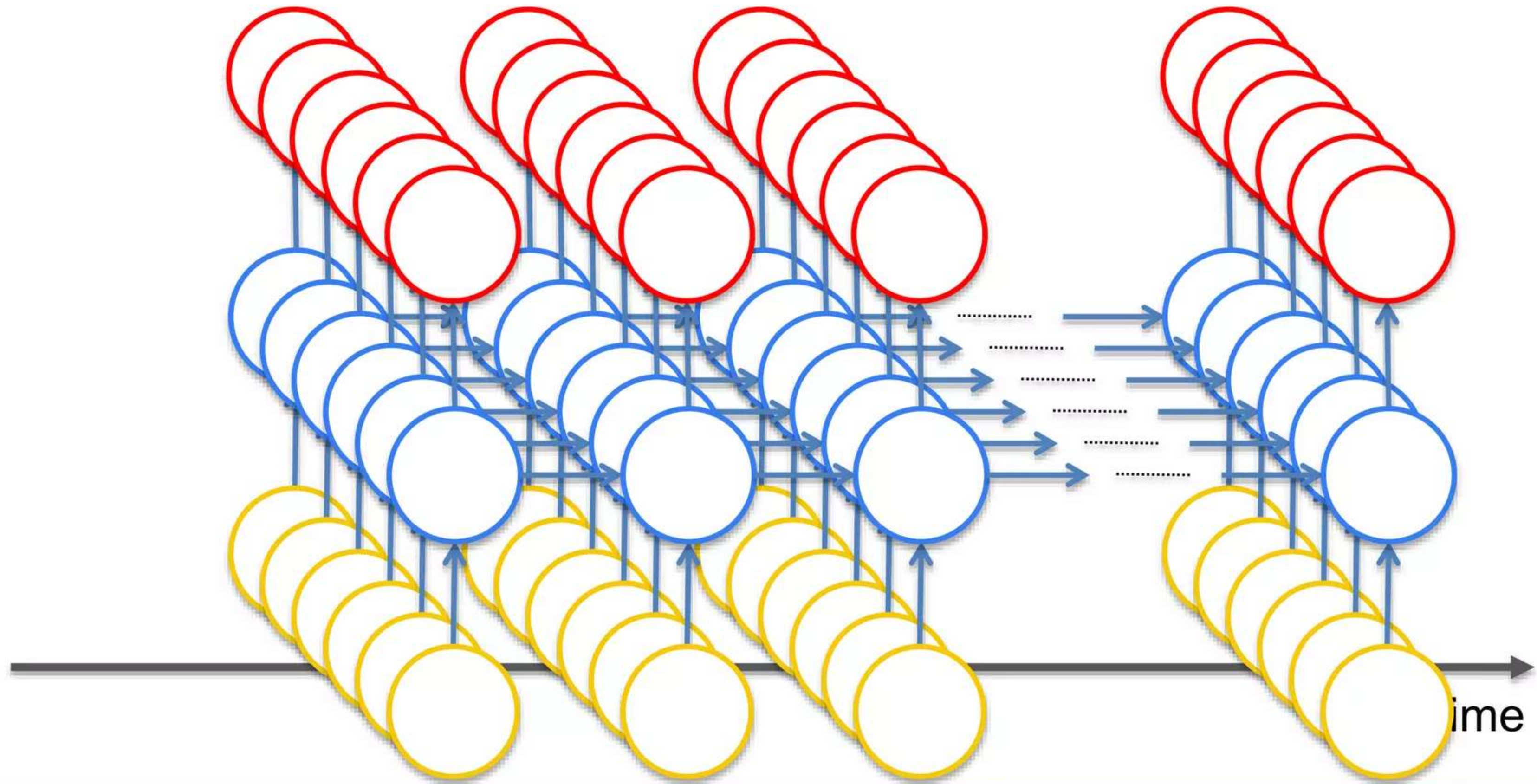
Recurrent Neural Networks



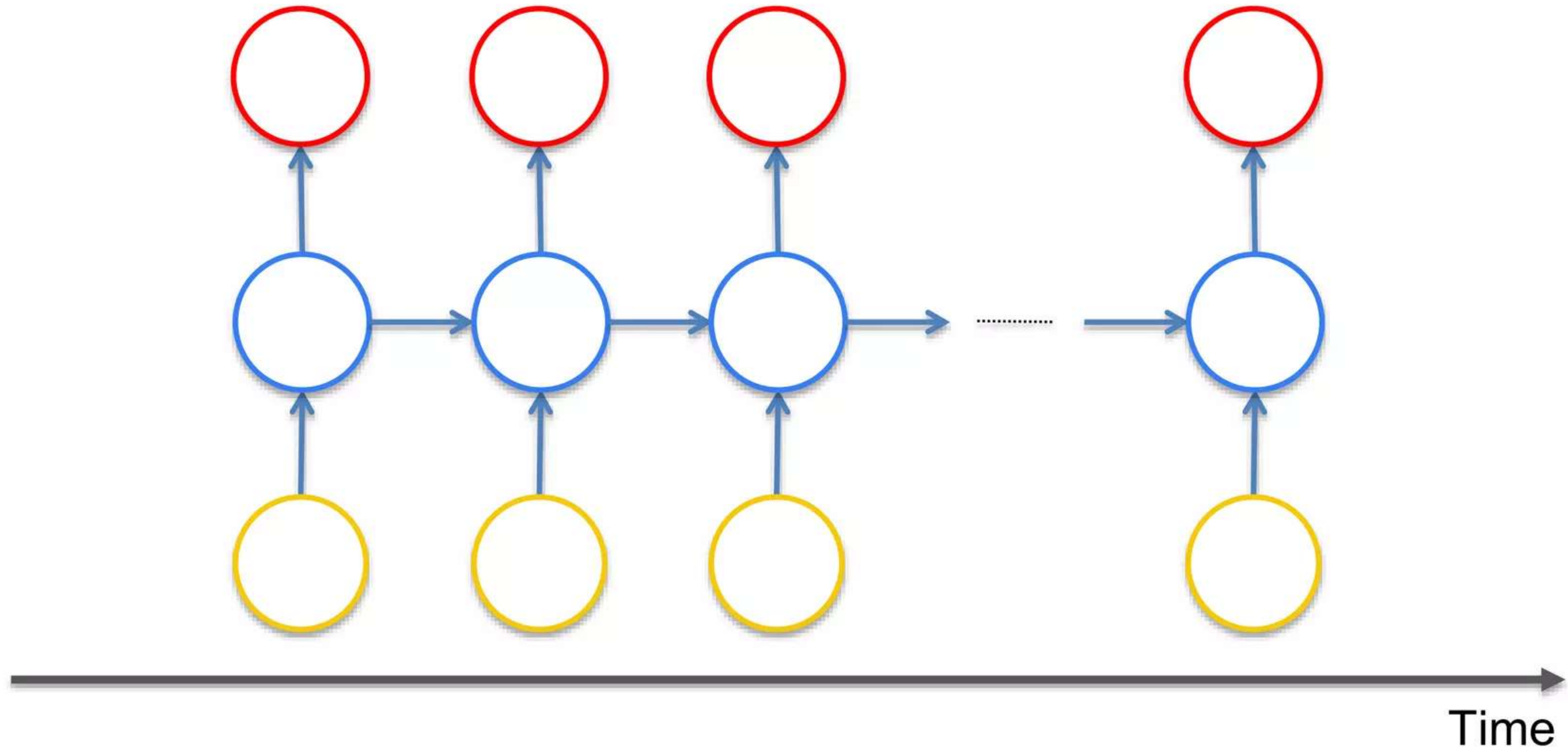
Recurrent Neural Networks



Recurrent Neural Networks



Recurrent Neural Networks

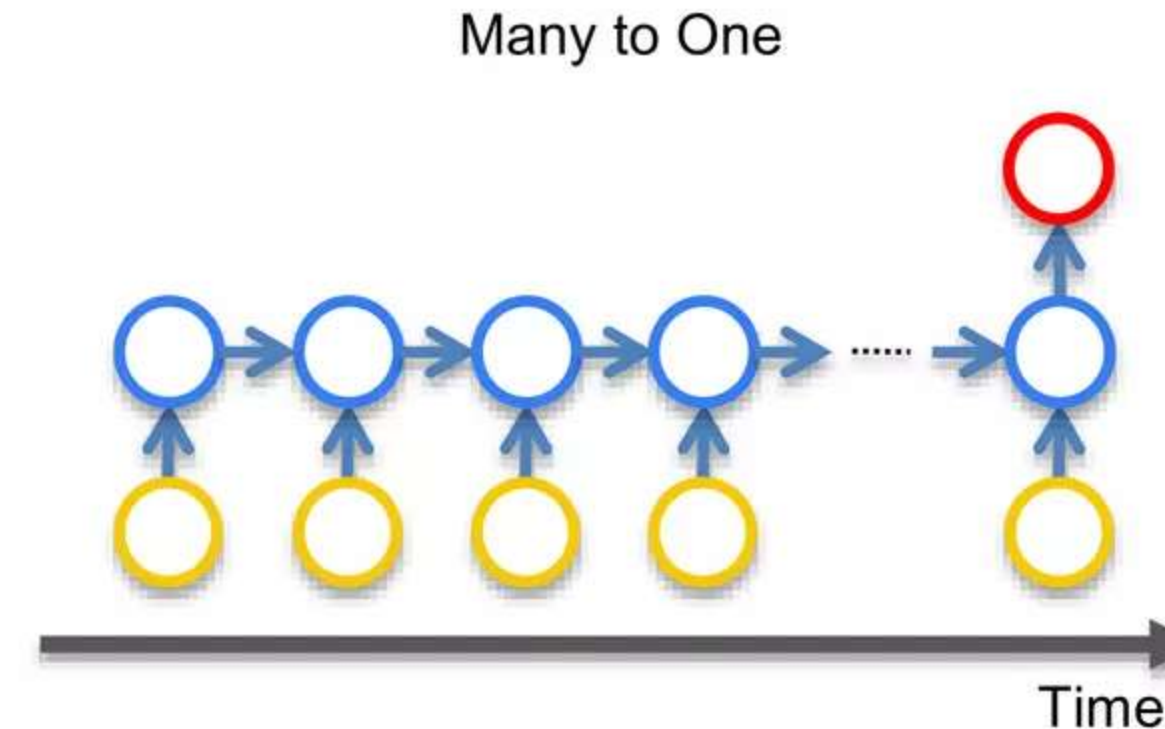
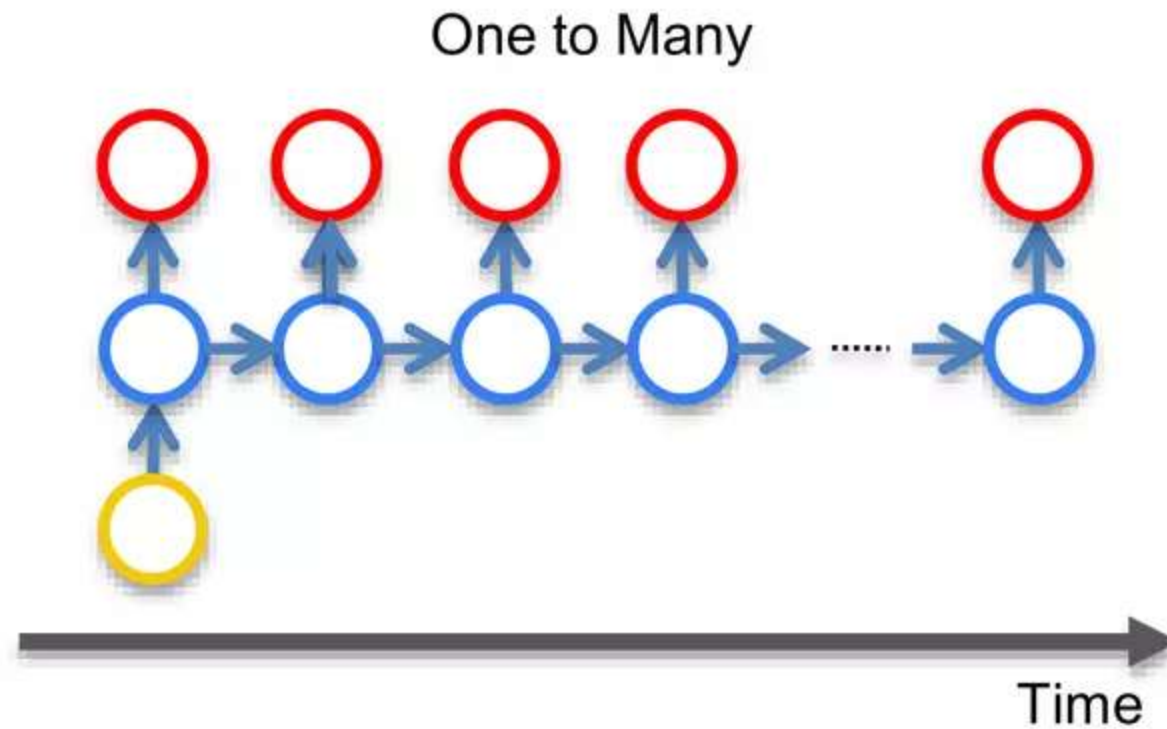


Recurrent Neural Networks



"black and white
dog jumps over
bar."

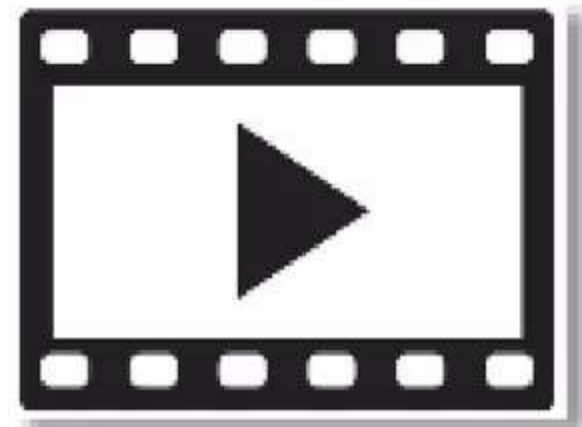
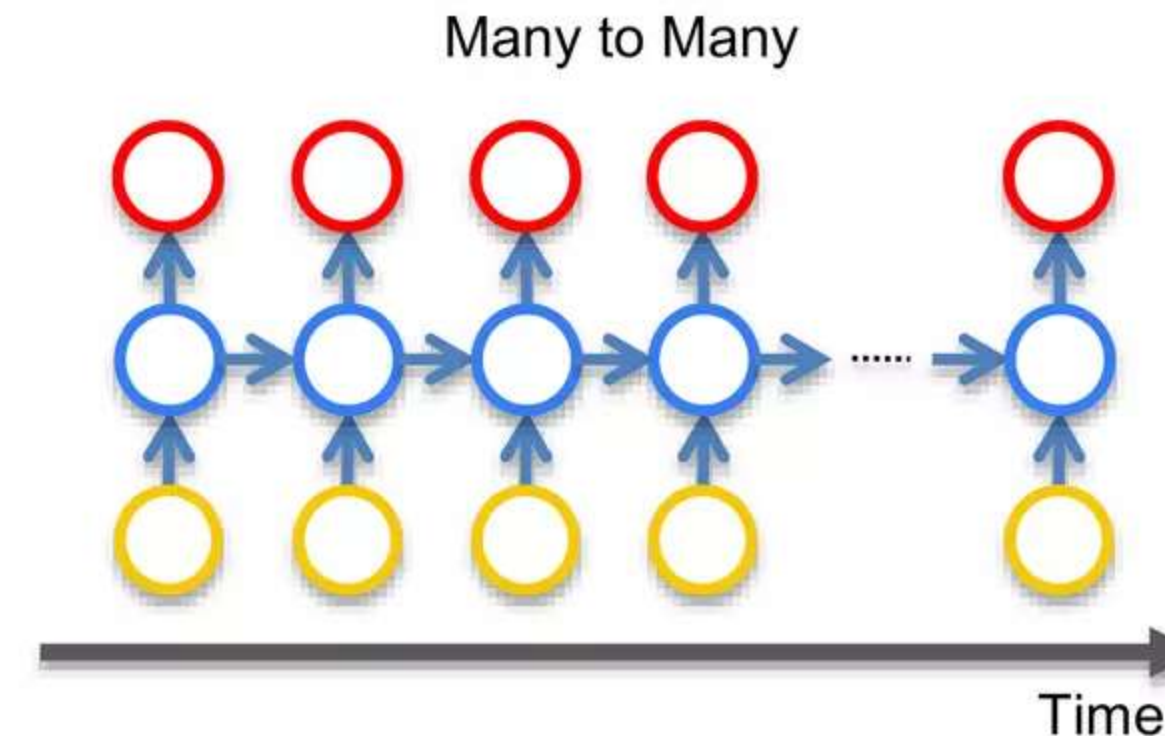
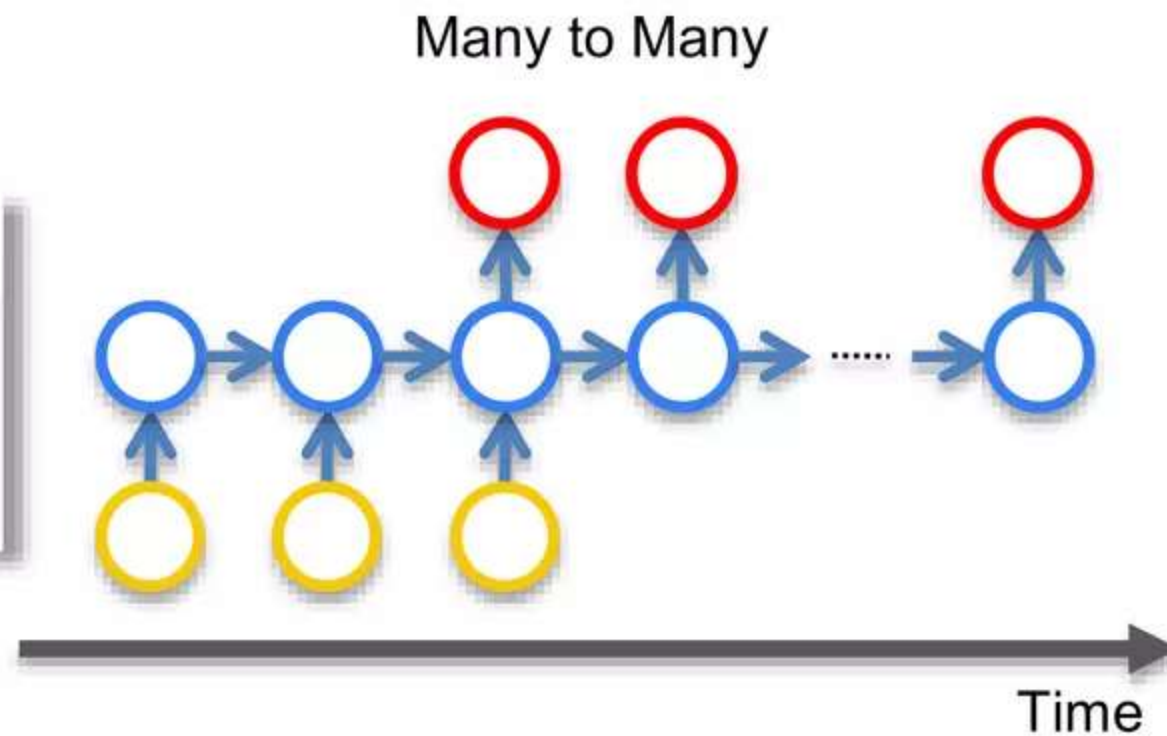
karpathy.github.io



"Thanks for a great
party at the
weekend, we really
enjoyed it!"

> sentiment: positive
score: 86%

dev.havenondemand.com



Reference: karpathy.github.io

The Activation Function

Additional Reading:

Sunspring (movie, 2016)

- Directed By Oscar Sharp
- Written by Benjamin

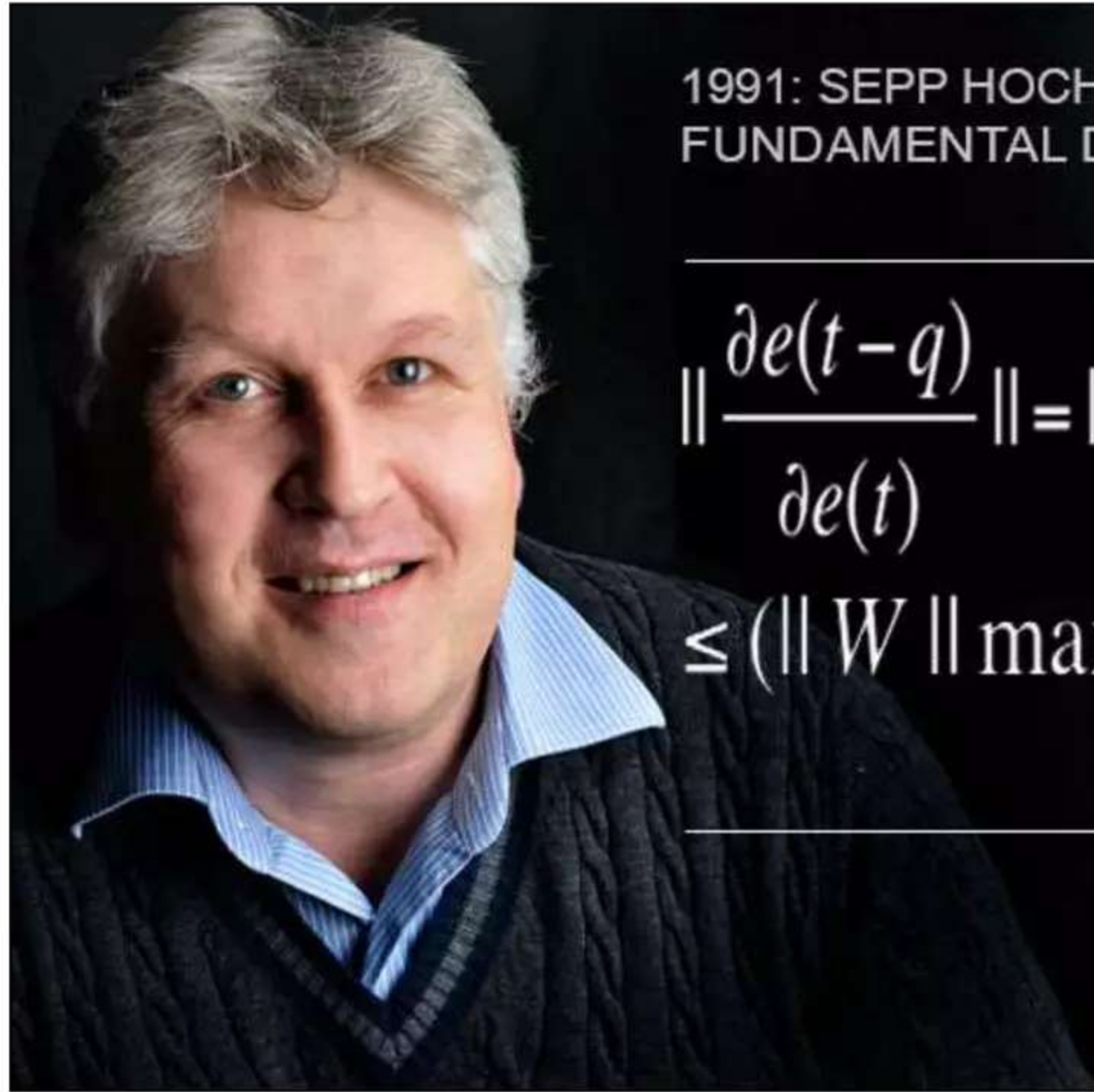


Link:

<https://arstechnica.com/the-multiverse/2016/06/an-ai-wrote-this-movie-and-its-strangely-moving/>

Vanishing Gradient

The Vanishing Gradient Problem

A portrait of Sepp Hochreiter, a man with grey hair, wearing a blue shirt and a dark sweater, smiling.

1991: SEPP HOCHREITER'S ANALYSIS OF THE
FUNDAMENTAL DEEP LEARNING PROBLEM

$$\left\| \frac{\partial e(t-q)}{\partial e(t)} \right\| = \left\| \prod_{m=1}^q W F'(Net(t-m)) \right\|$$
$$\leq (\|W\| \max_{Net} \{ \|F'(Net)\| \})^q$$

Image Source: people.idsia.ch

The Vanishing Gradient Problem

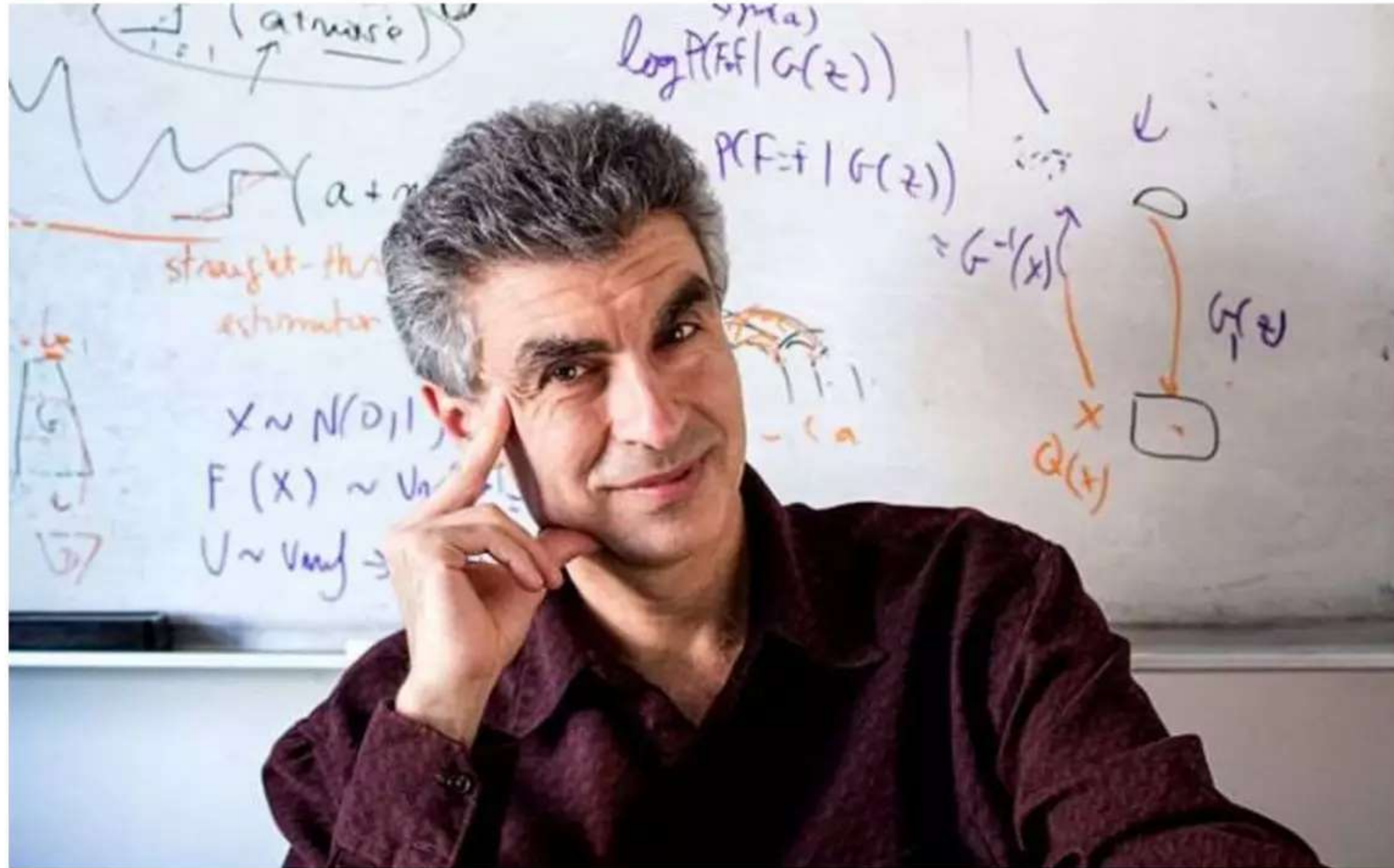


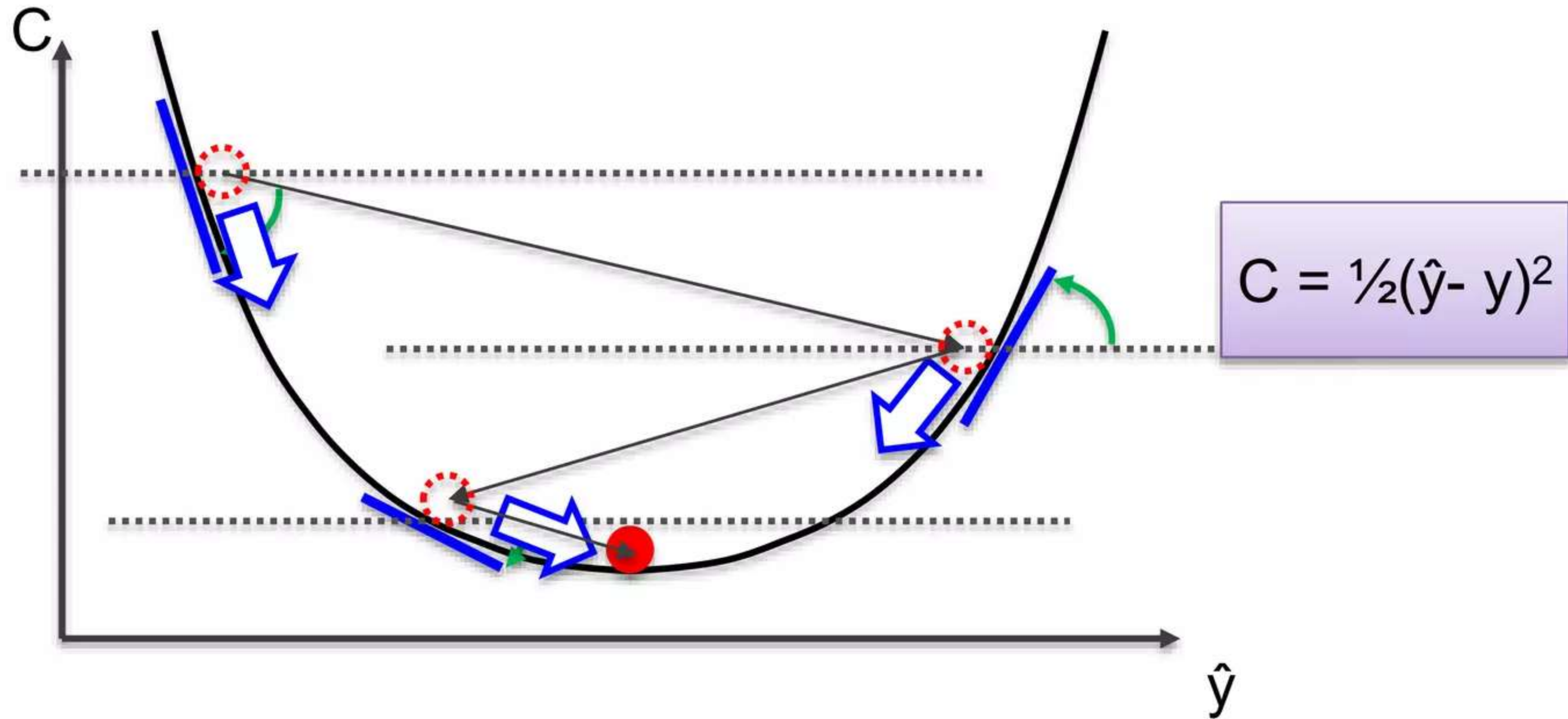
Image Source: Université Montréal

The Vanishing Gradient Problem

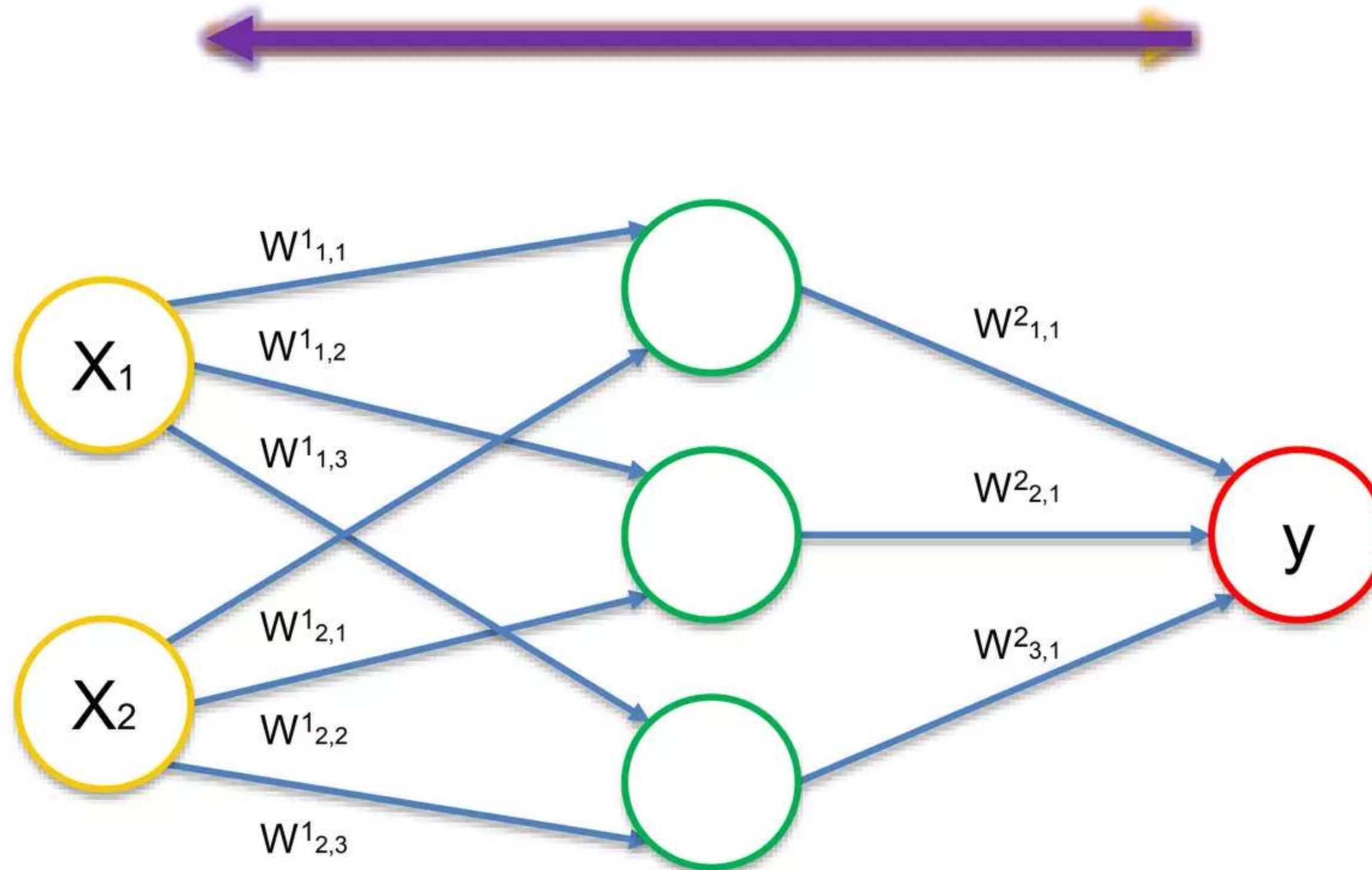


Image Source: recode.net

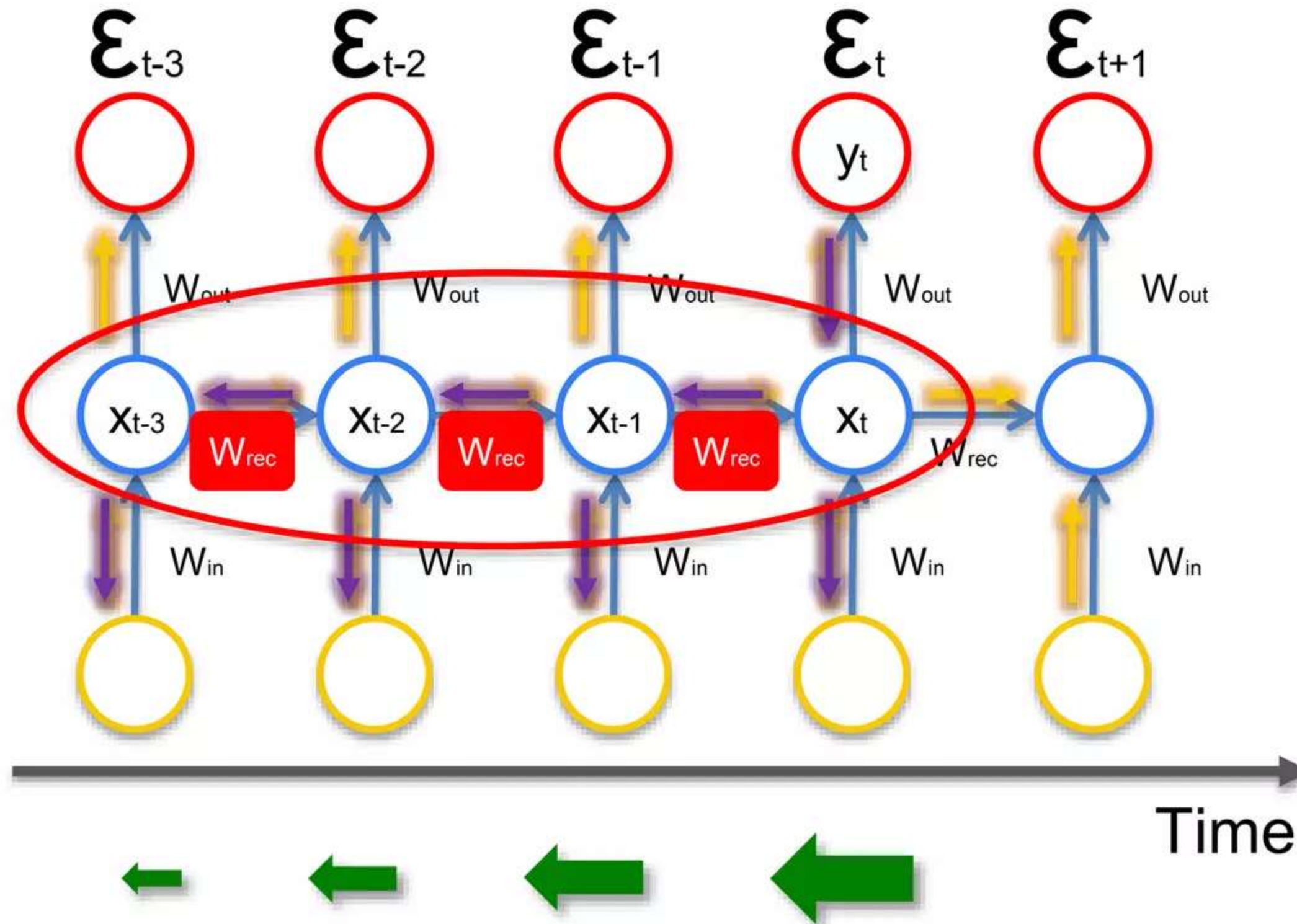
The Vanishing Gradient Problem



The Vanishing Gradient Problem



The Vanishing Gradient Problem



$$\frac{\partial \mathcal{E}}{\partial \theta} = \sum_{1 \leq t \leq T} \frac{\partial \mathcal{E}_t}{\partial \theta} \quad (3)$$

$$\frac{\partial \mathcal{E}_t}{\partial \theta} = \sum_{1 \leq k \leq t} \left(\frac{\partial \mathcal{E}_t}{\partial \mathbf{x}_t} \frac{\partial \mathbf{x}_t}{\partial \mathbf{x}_k} \frac{\partial^+ \mathbf{x}_k}{\partial \theta} \right) \quad (4)$$

$$\frac{\partial \mathbf{x}_t}{\partial \mathbf{x}_k} = \prod_{t \geq i > k} \frac{\partial \mathbf{x}_i}{\partial \mathbf{x}_{i-1}} = \prod_{t \geq i > k} \mathbf{W}_{rec}^T \text{diag}(\sigma'(\mathbf{x}_{i-1})) \quad (5)$$

$W_{rec} \sim \text{small}$	$\Rightarrow$	Vanishing
$W_{rec} \sim \text{large}$	$\Rightarrow$	Exploding

Formula Source: Razvan Pascanu et al. (2013)

The Vanishing Gradient Problem

Solutions:

1. Exploding Gradient

- Truncated Backpropagation
- Penalties
- Gradient Clipping

2. Vanishing Gradient

- Weight Initialization
- Echo State Networks
- Long Short-Term Memory Networks (LSTMs)



The Vanishing Gradient Problem

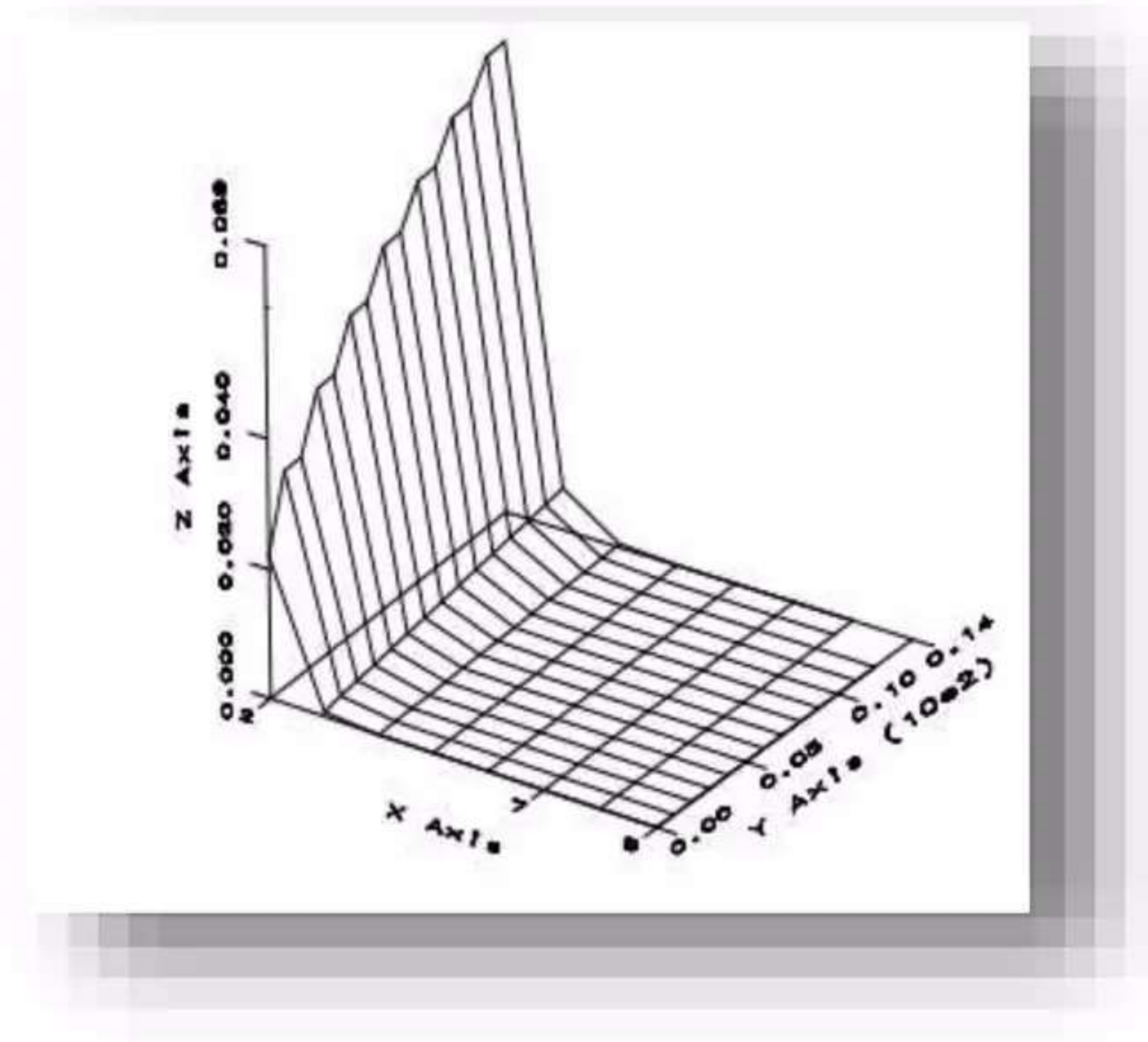
Additional Reading:

Untersuchungen zu dynamischen neuronalen Netzen

By Sepp (Josef) Hochreiter (1991)

Link:

<http://people.idsia.ch/~juergen/SeppHochreiter1991ThesisAdvisorSchmidhuber.pdf>



The Vanishing Gradient Problem

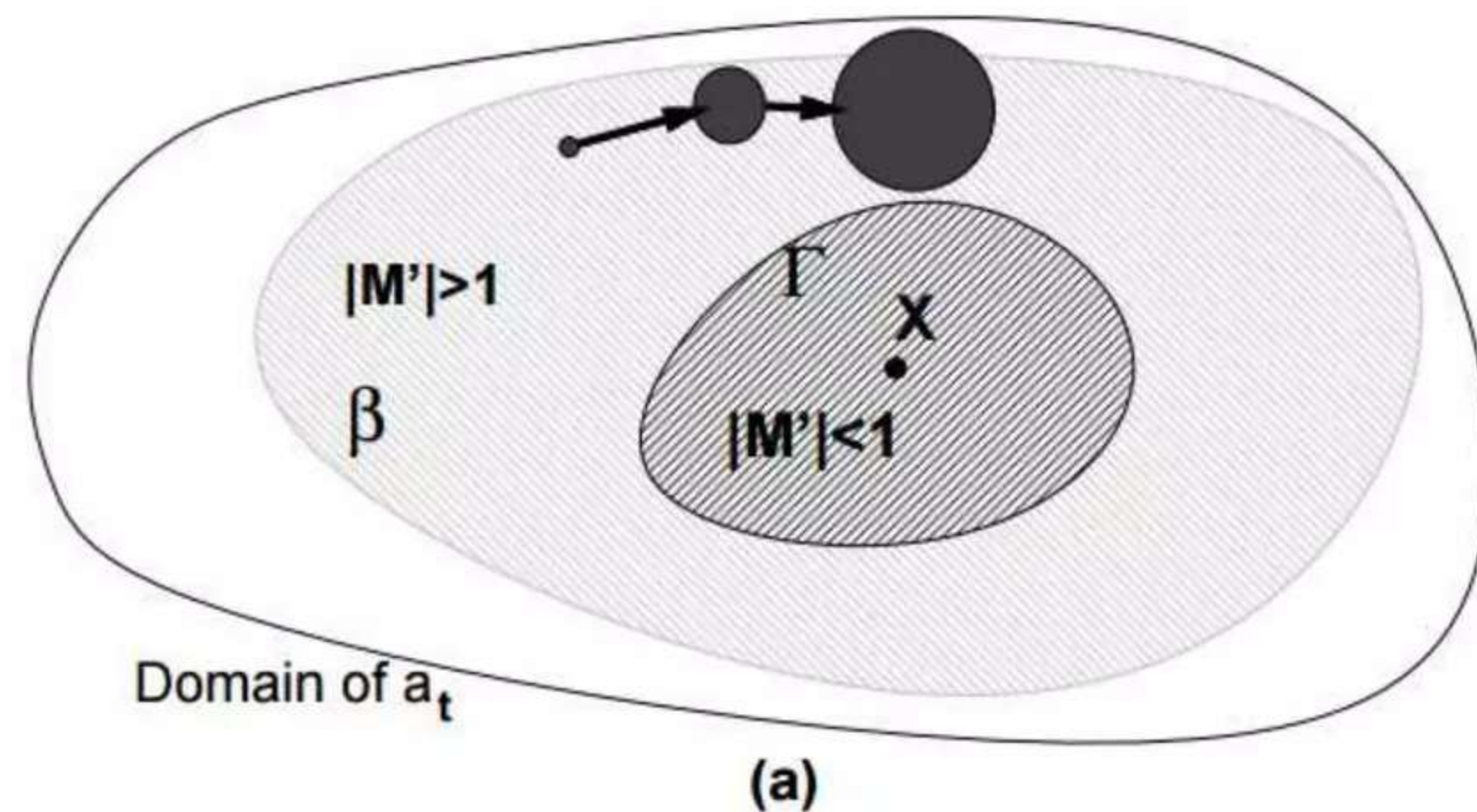
Additional Reading:

Learning Long-Term Dependencies with Gradient Descent is Difficult

By Yoshua Bengio et al. (1994)

Link:

<http://www-dsi.ing.unifi.it/~paolo/ps/tnn-94-gradient.pdf>



The Vanishing Gradient Problem

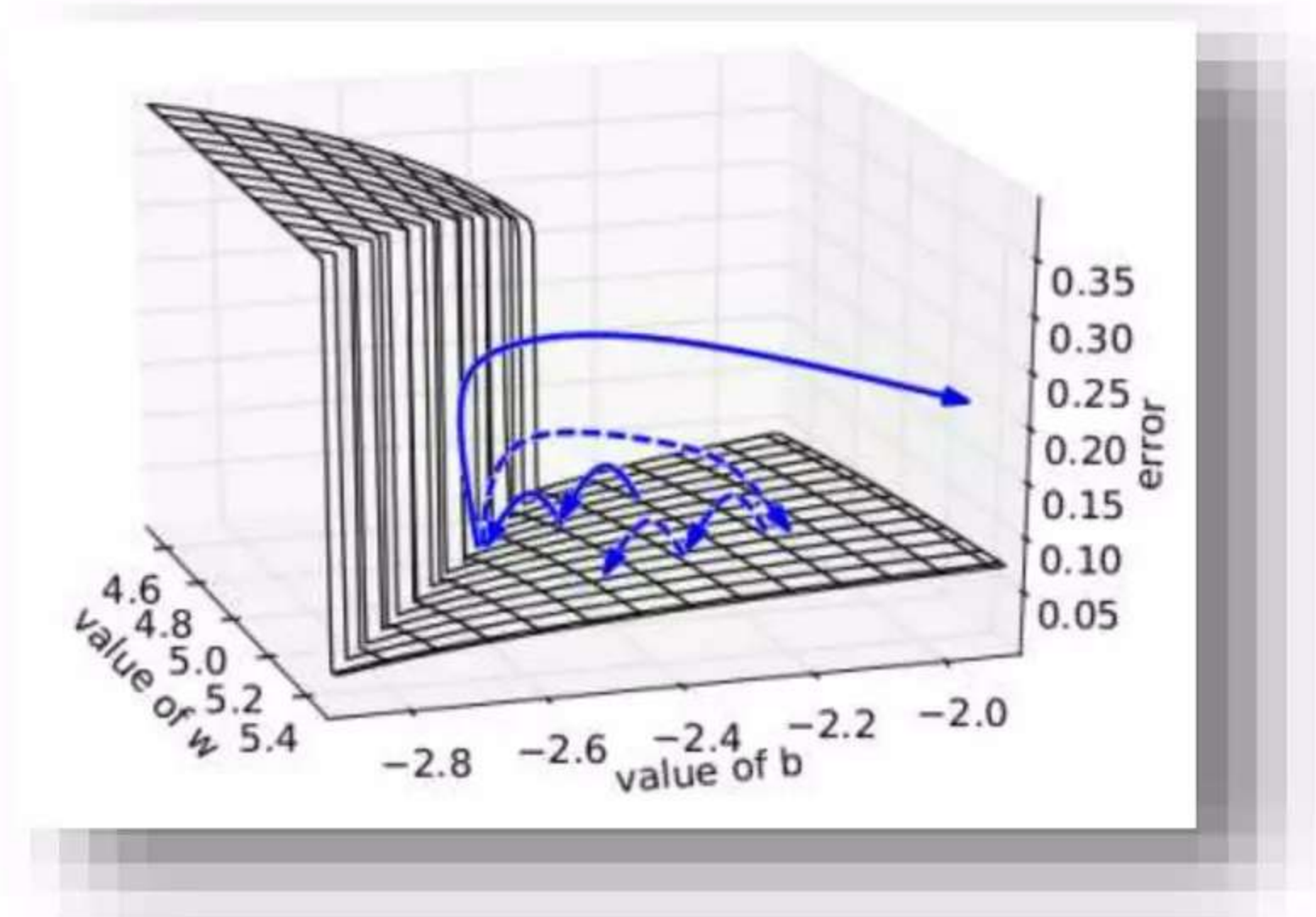
Additional Reading:

On the difficulty of training recurrent neural networks

By Razvan Pascanu et al. (2013)

Link:

<http://www.jmlr.org/proceedings/papers/v28/pascanu13.pdf>



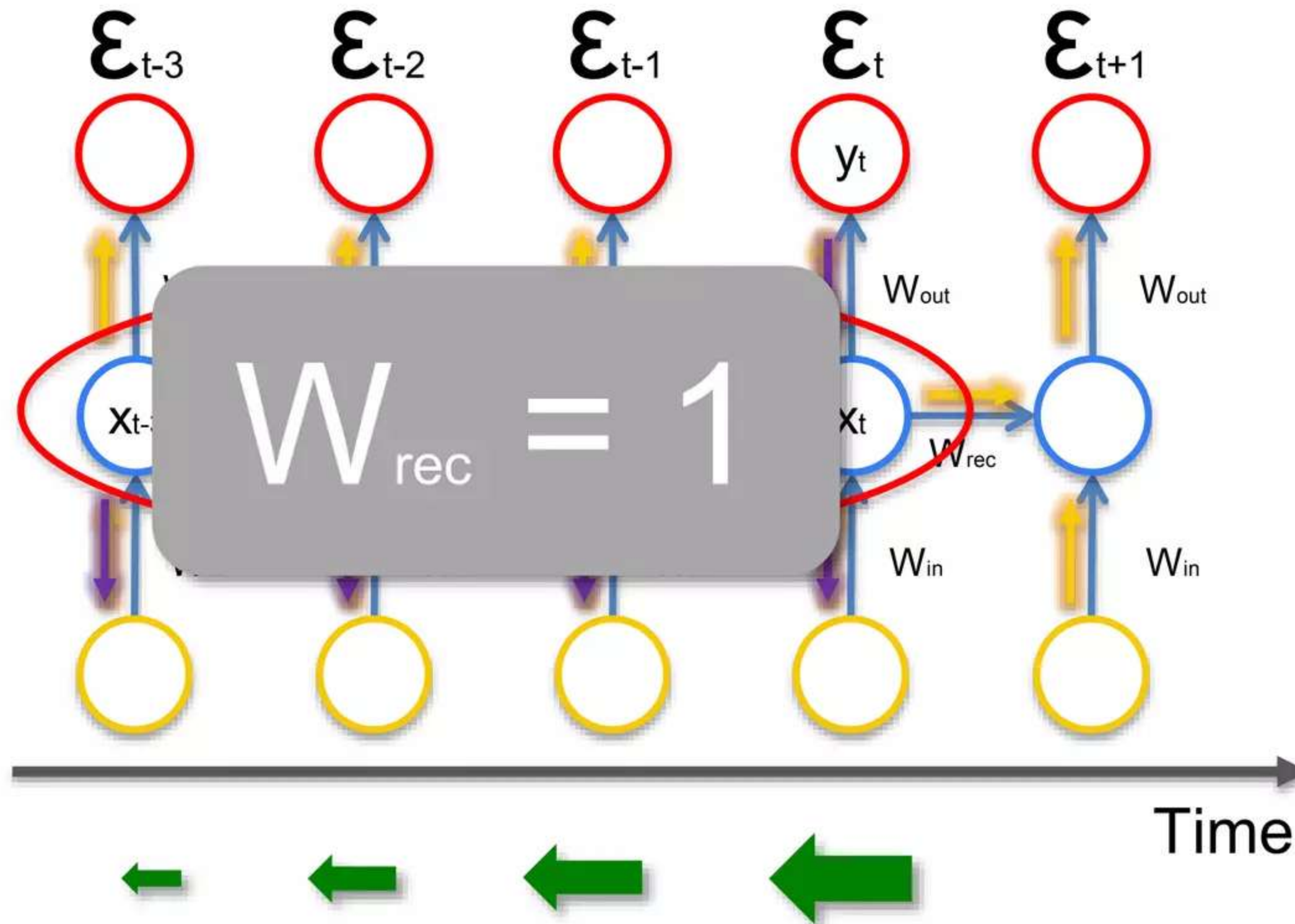
LSTMs

Long Short-Term Memory

Today:

- A bit of history
- LSTM Architecture
- Example walkthrough

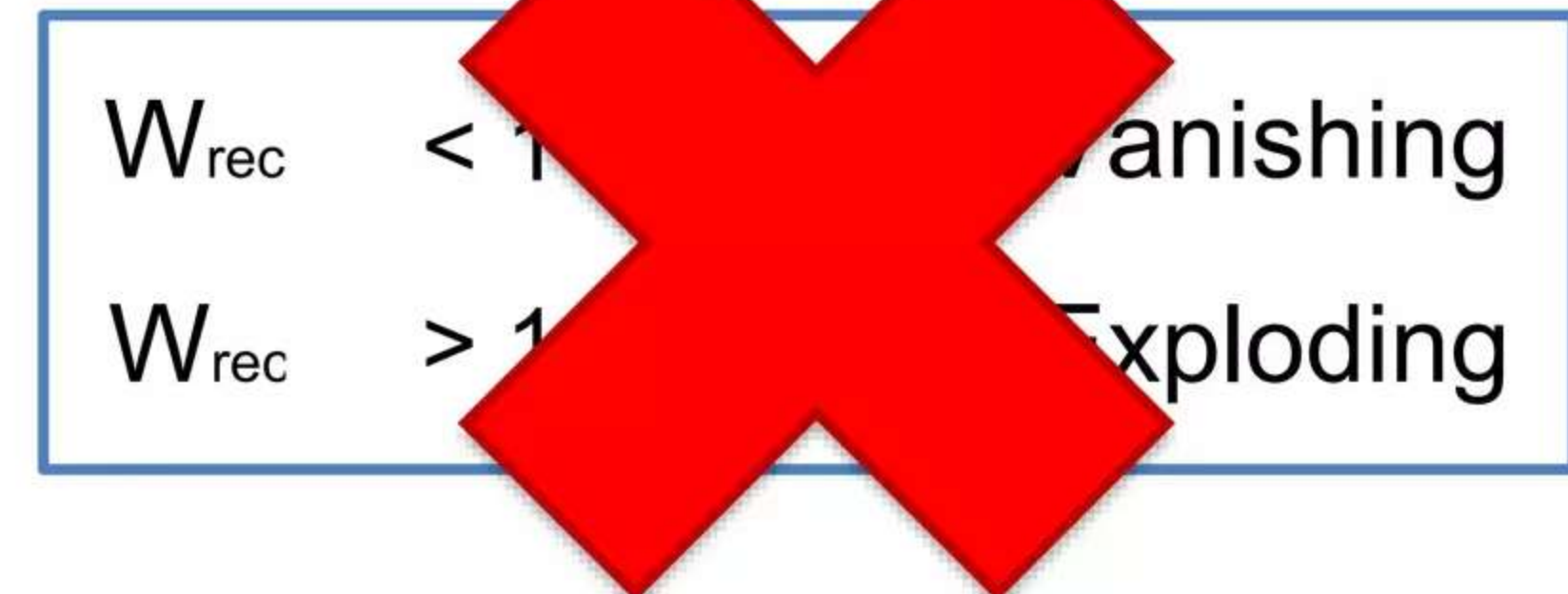
Long Short-Term Memory



$$\frac{\partial \mathcal{E}}{\partial \theta} = \sum_{1 \leq t \leq T} \frac{\partial \mathcal{E}_t}{\partial \theta} \quad (3)$$

$$\frac{\partial \mathcal{E}_t}{\partial \theta} = \sum_{1 \leq k \leq t} \left(\frac{\partial \mathcal{E}_t}{\partial \mathbf{x}_t} \frac{\partial \mathbf{x}_t}{\partial \mathbf{x}_k} \frac{\partial^+ \mathbf{x}_k}{\partial \theta} \right) \quad (4)$$

$$\frac{\partial \mathbf{x}_t}{\partial \mathbf{x}_k} = \prod_{t \geq i > k} \frac{\partial \mathbf{x}_i}{\partial \mathbf{x}_{i-1}} = \prod_{t \geq i > k} \mathbf{W}_{rec}^T \text{diag}(\sigma'(\mathbf{x}_{i-1})) \quad (5)$$



Formula Source: Razvan Pascanu et al. (2013)

Long Short-Term Memory

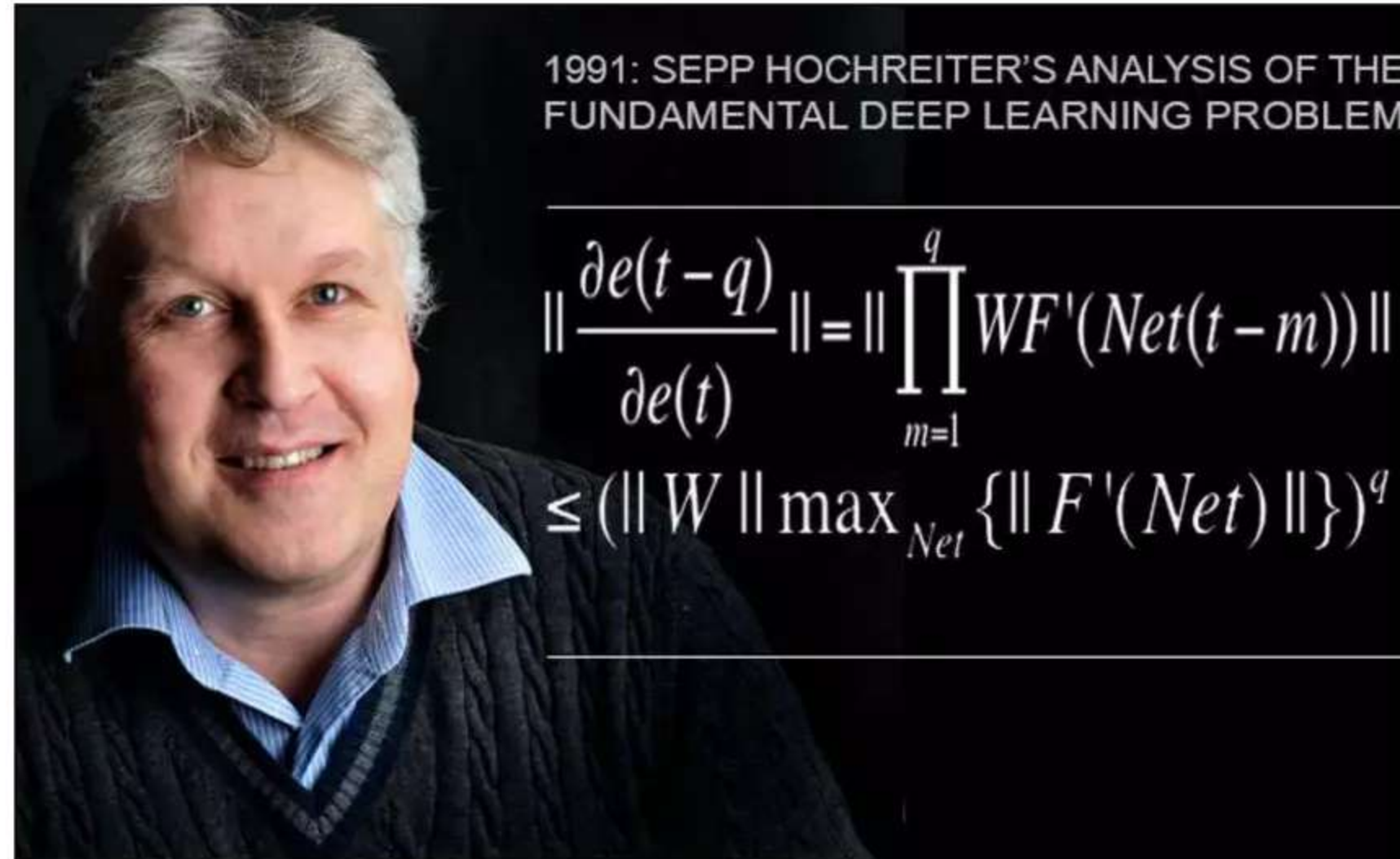
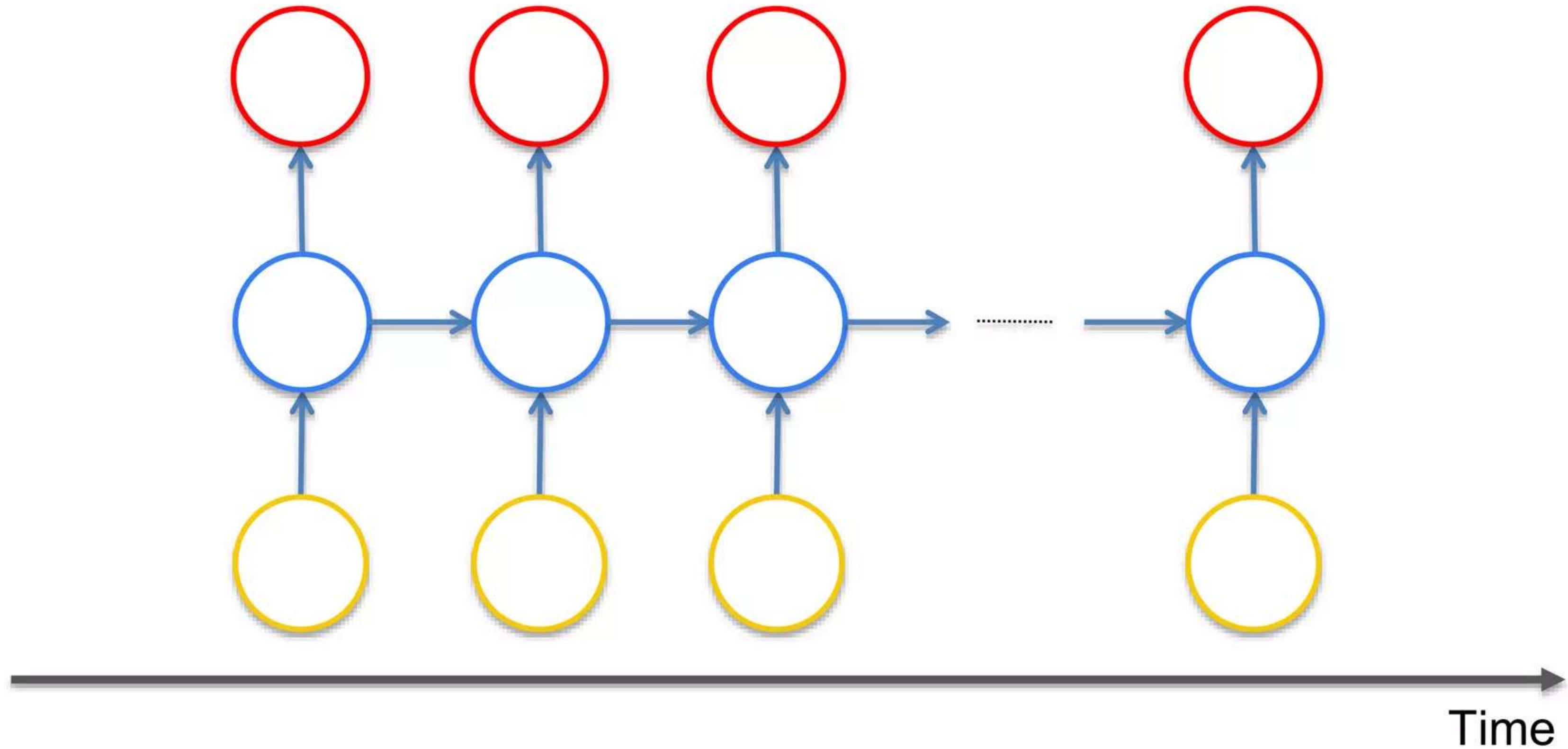


Image Sources: people.idsia.ch, ics.usi.ch

Long Short-Term Memory



Long Short-Term Memory

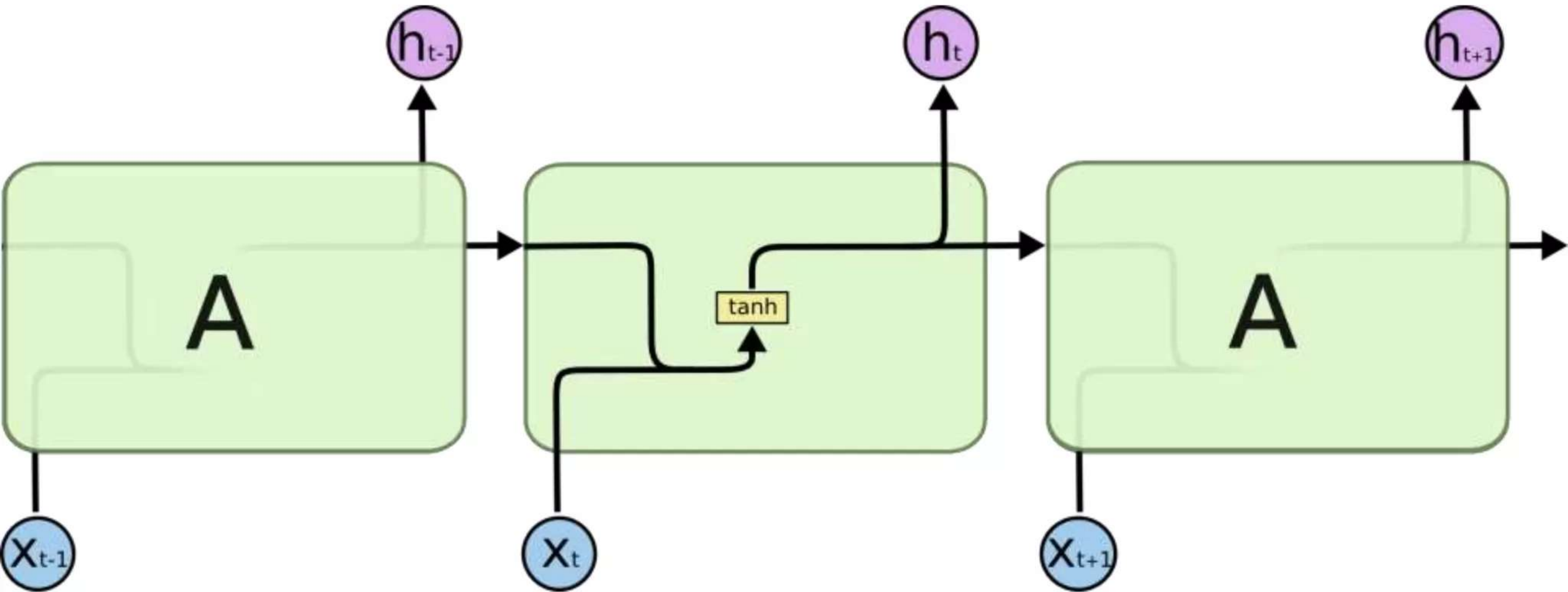


Image Source: colah.github.io

Long Short-Term Memory

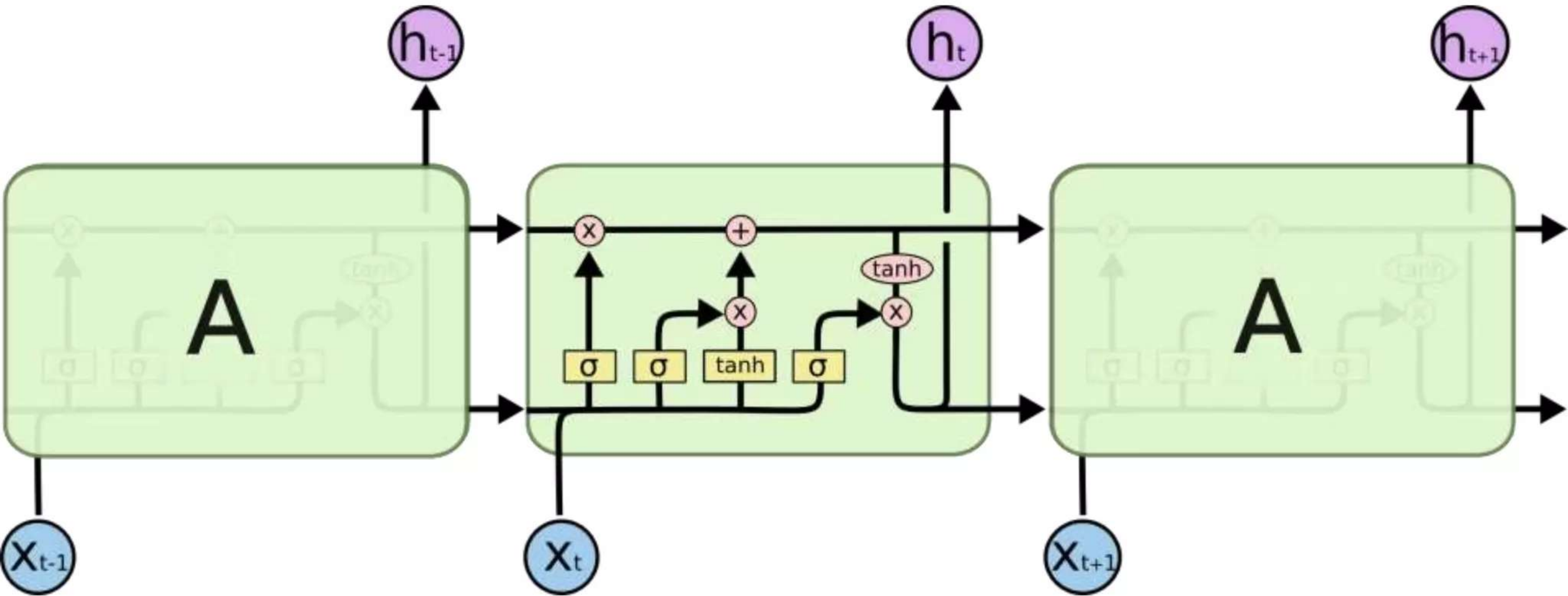


Image Source: colah.github.io

Long Short-Term Memory

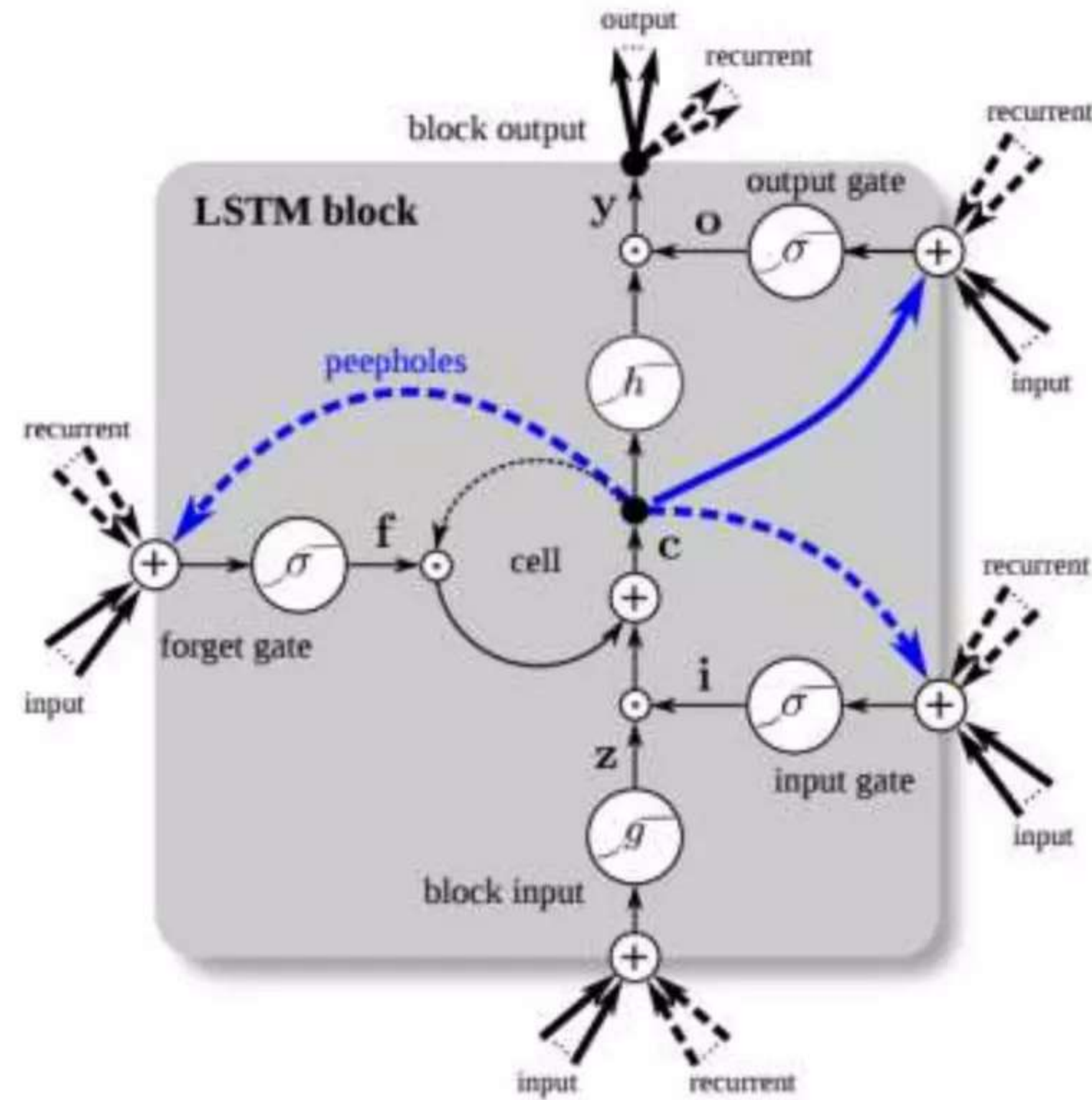


Image Source: arxiv.org/pdf/1503.04069.pdf

Long Short-Term Memory

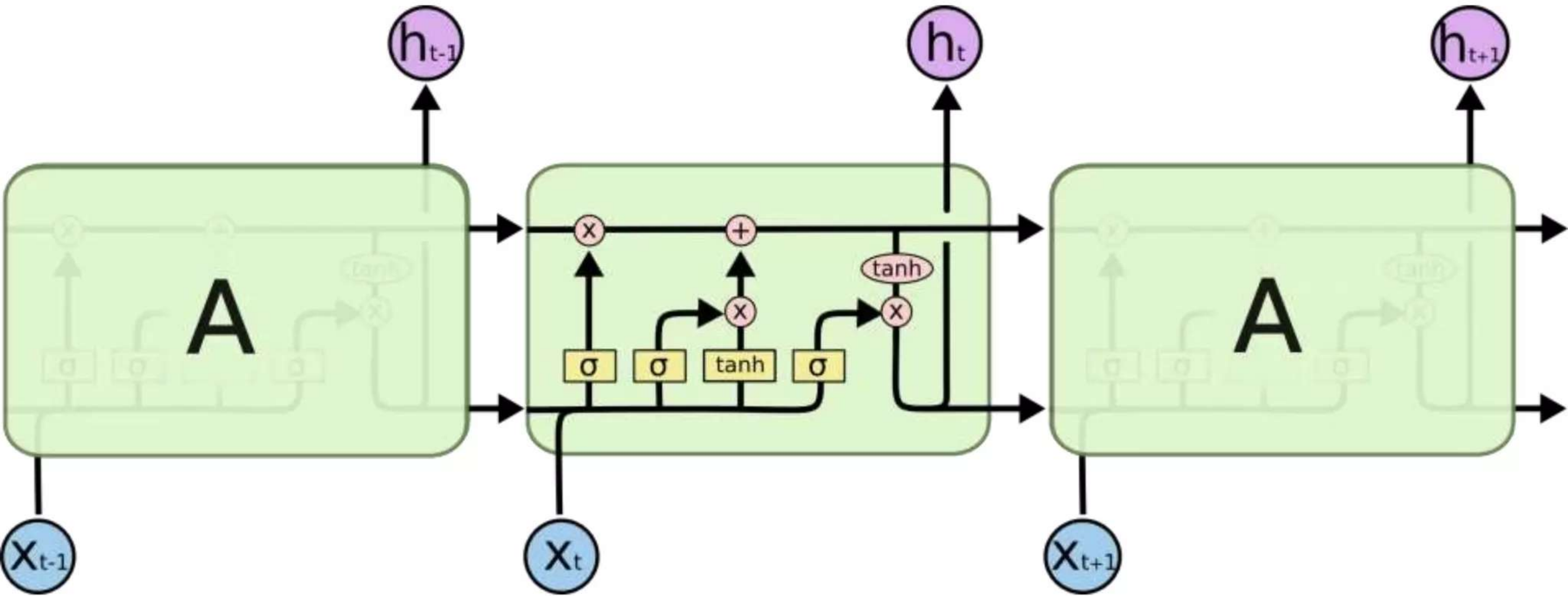


Image Source: colah.github.io

Long Short-Term Memory

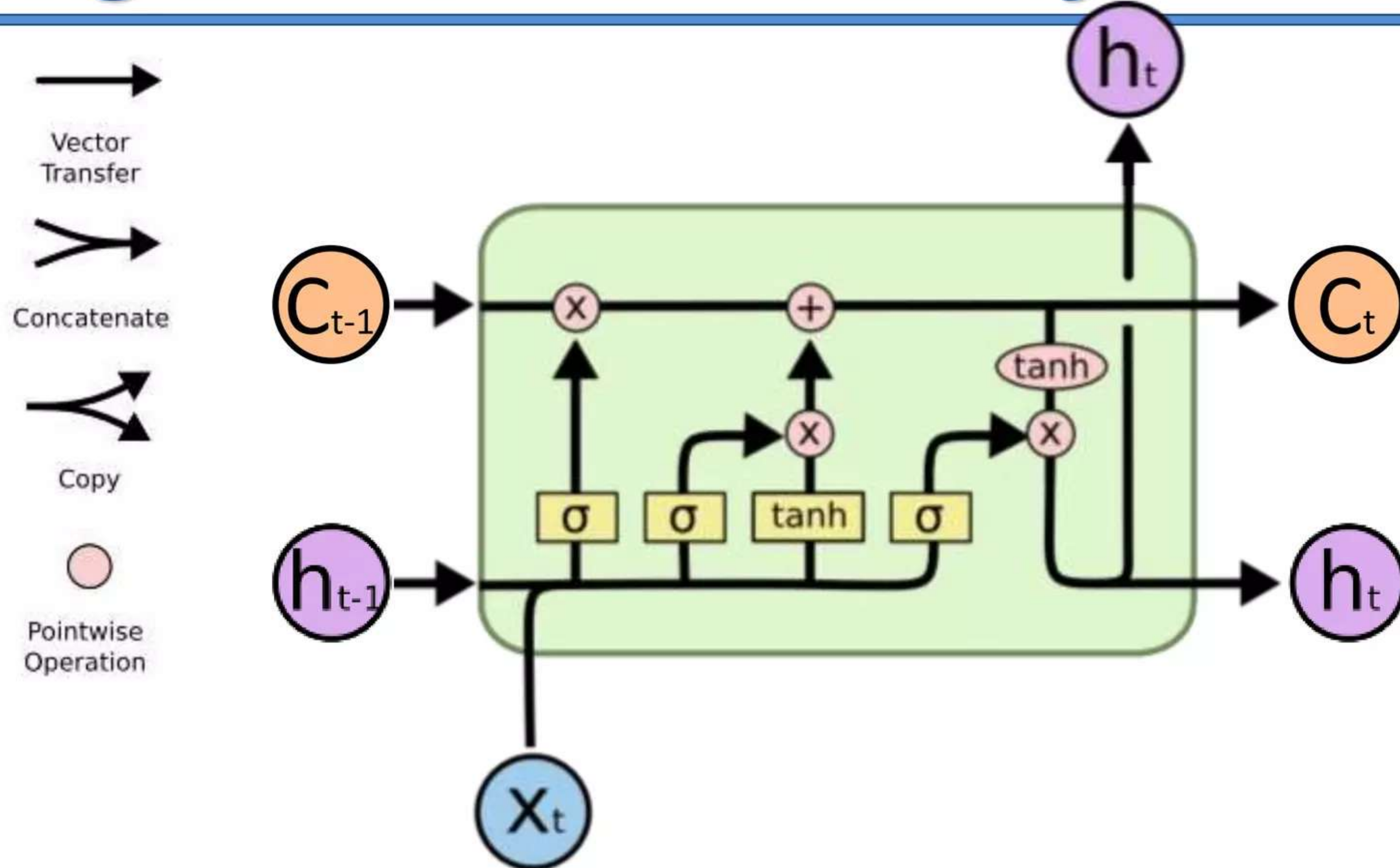


Image Source: colah.github.io

Long Short-Term Memory



Long Short-Term Memory

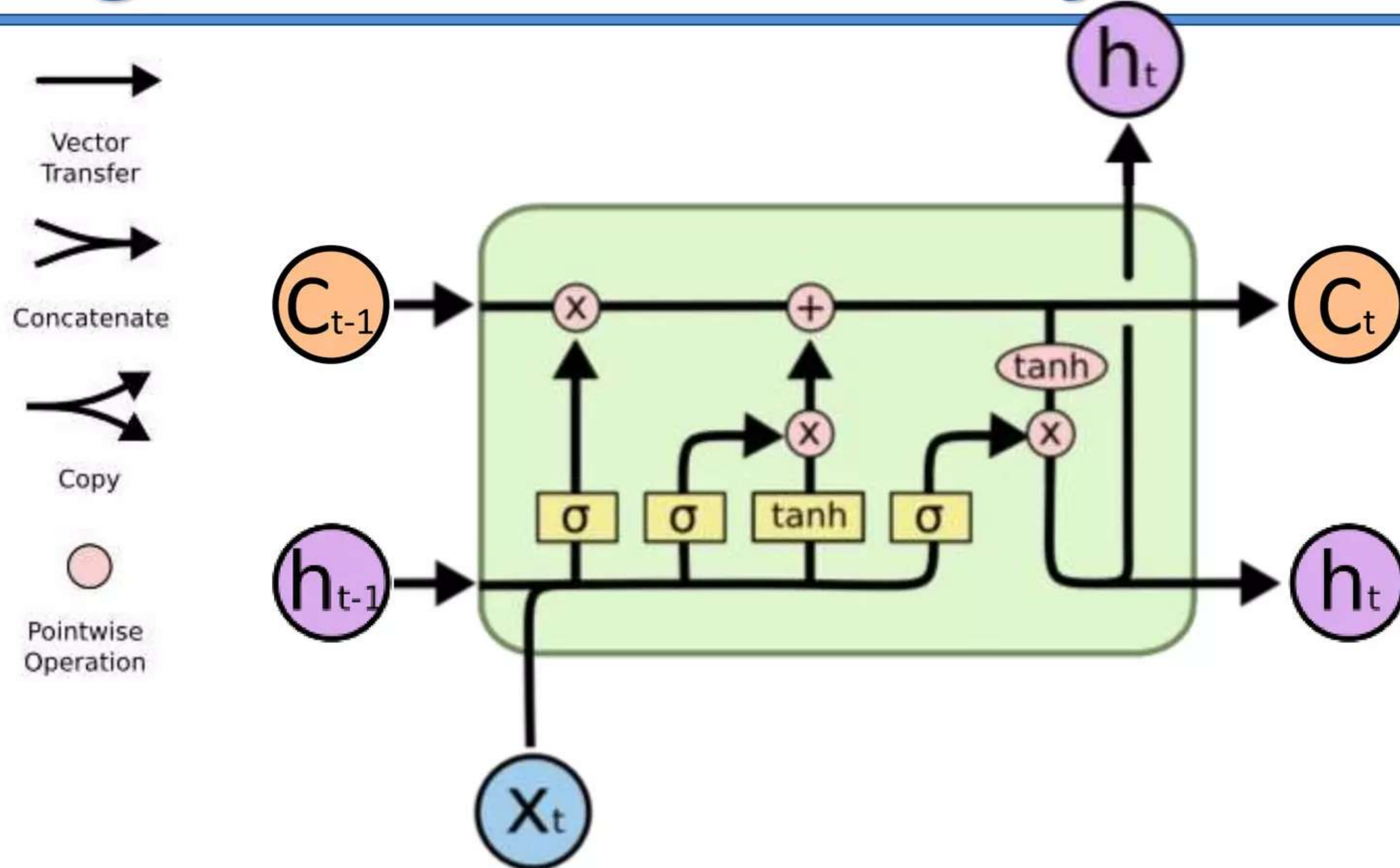


Image Source: colah.github.io

Long Short-Term Memory



Long Short-Term Memory

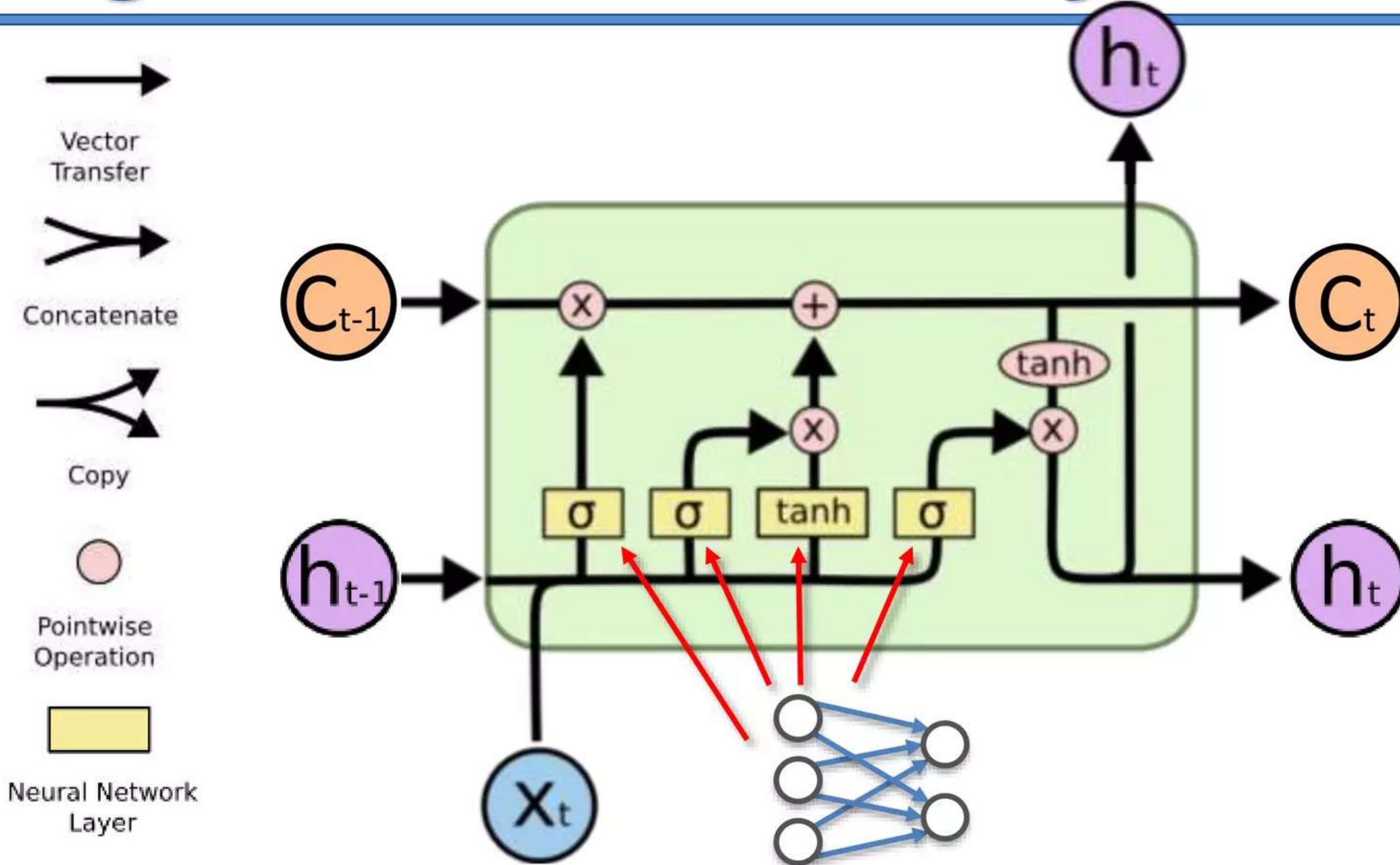


Image Source: colah.github.io

Long Short-Term Memory

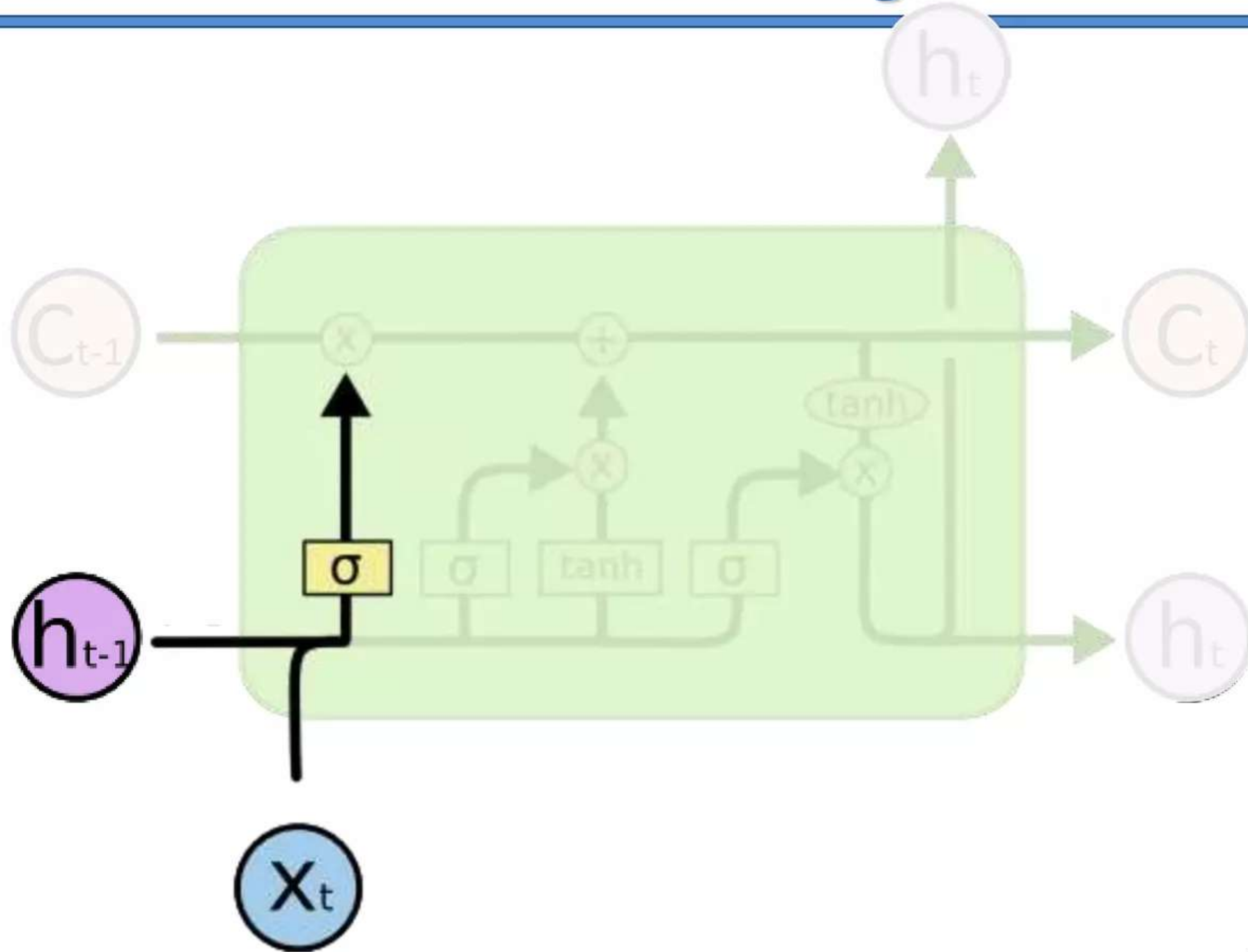


Image Source: colah.github.io

Long Short-Term Memory

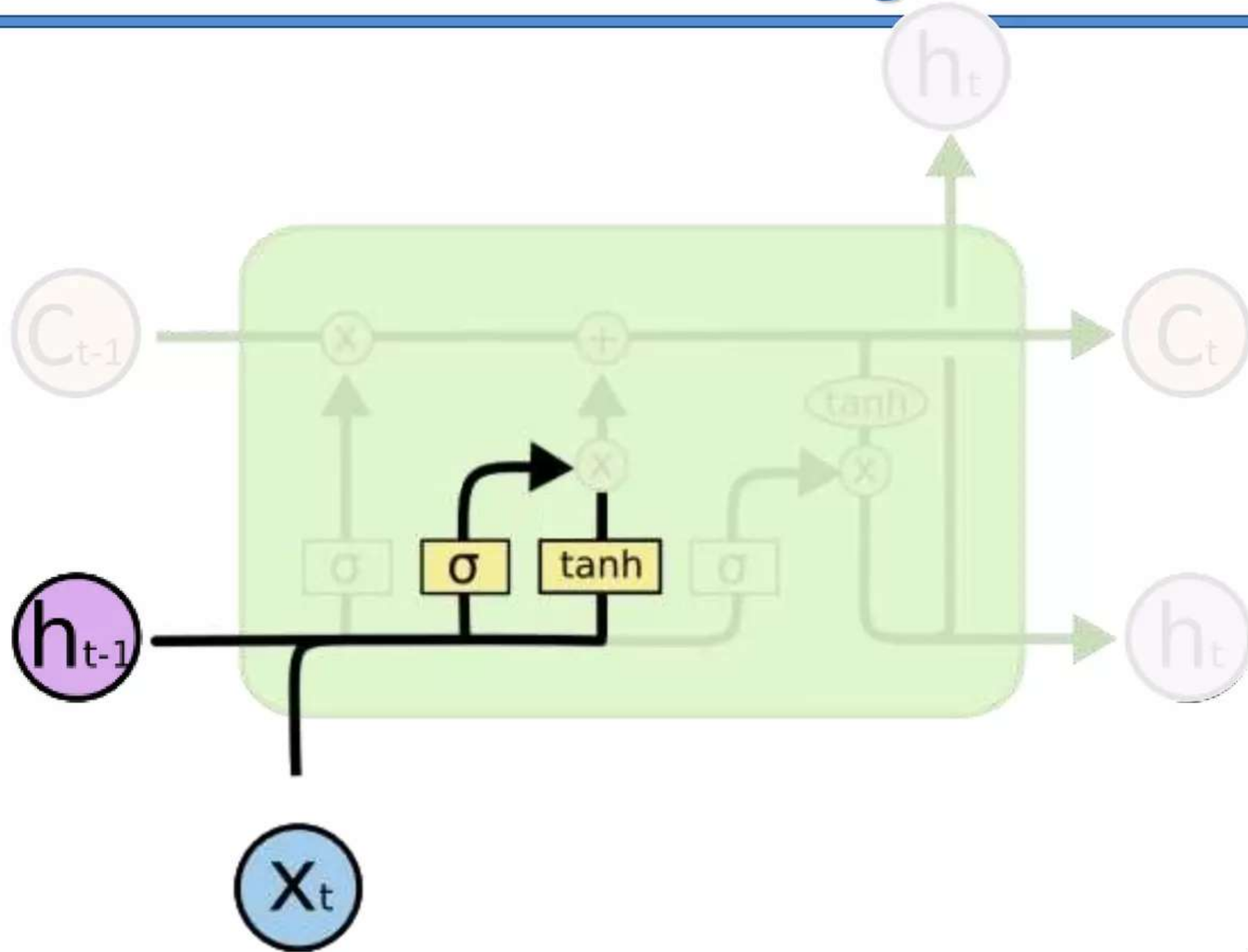


Image Source: colah.github.io

Long Short-Term Memory

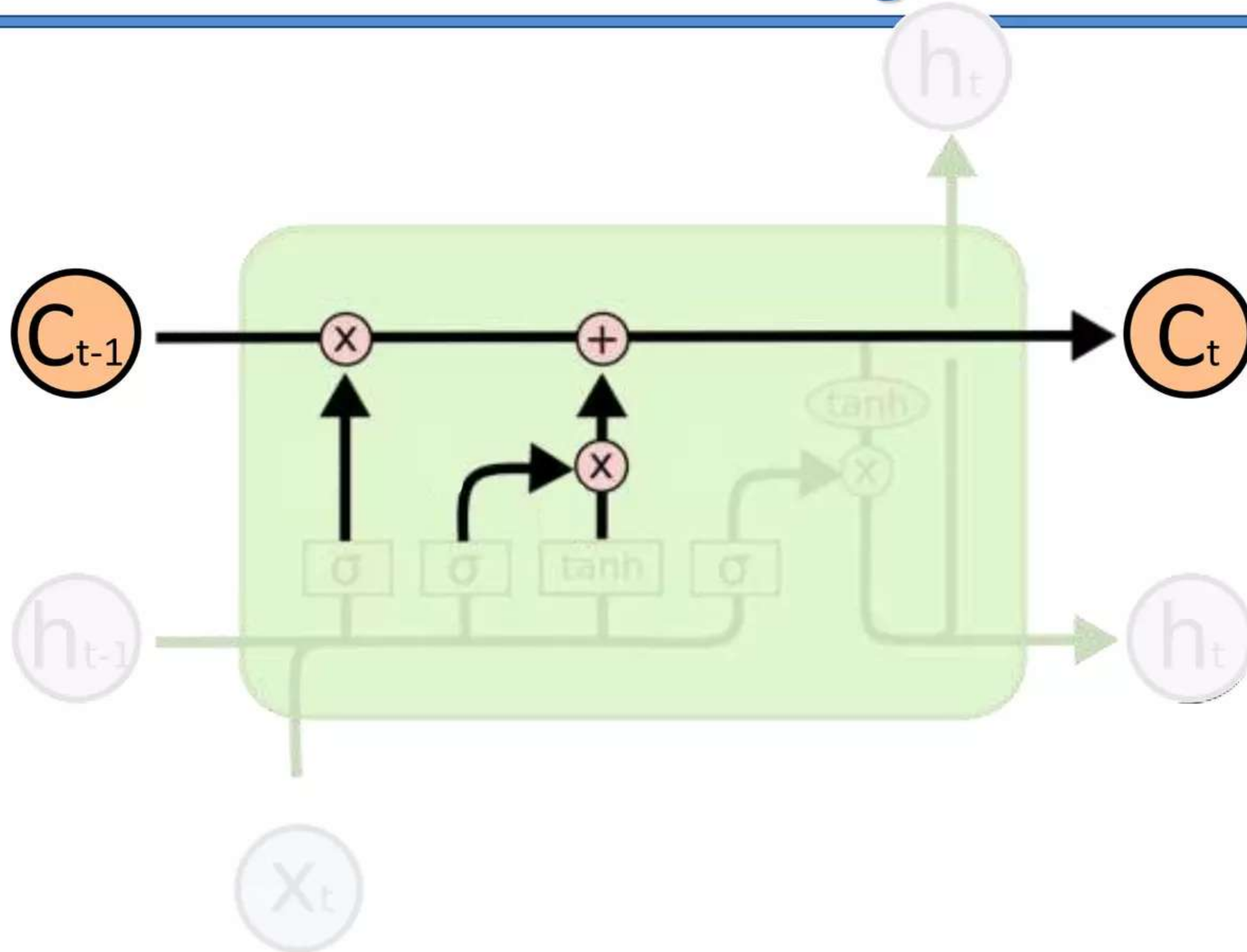


Image Source: colah.github.io

Long Short-Term Memory

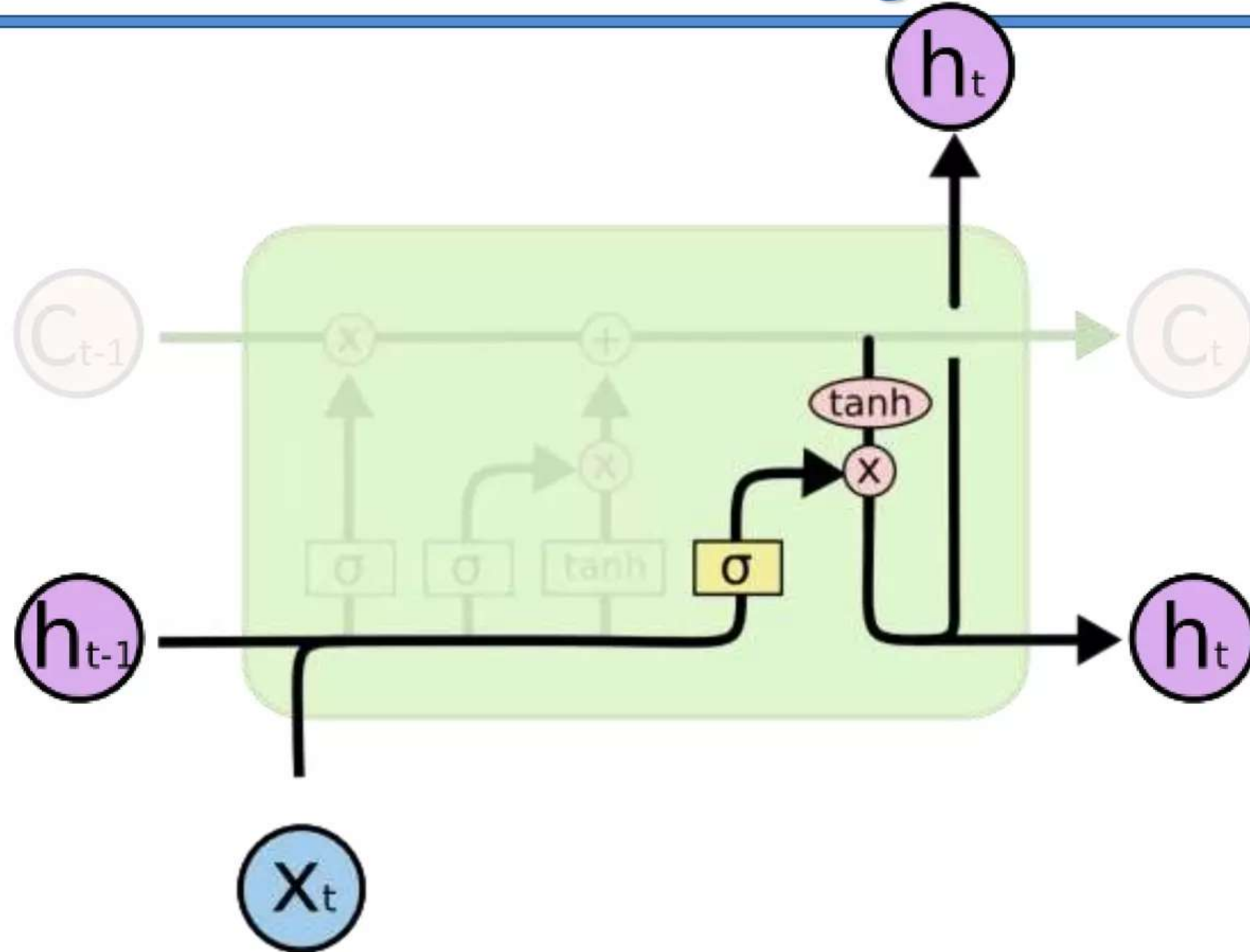
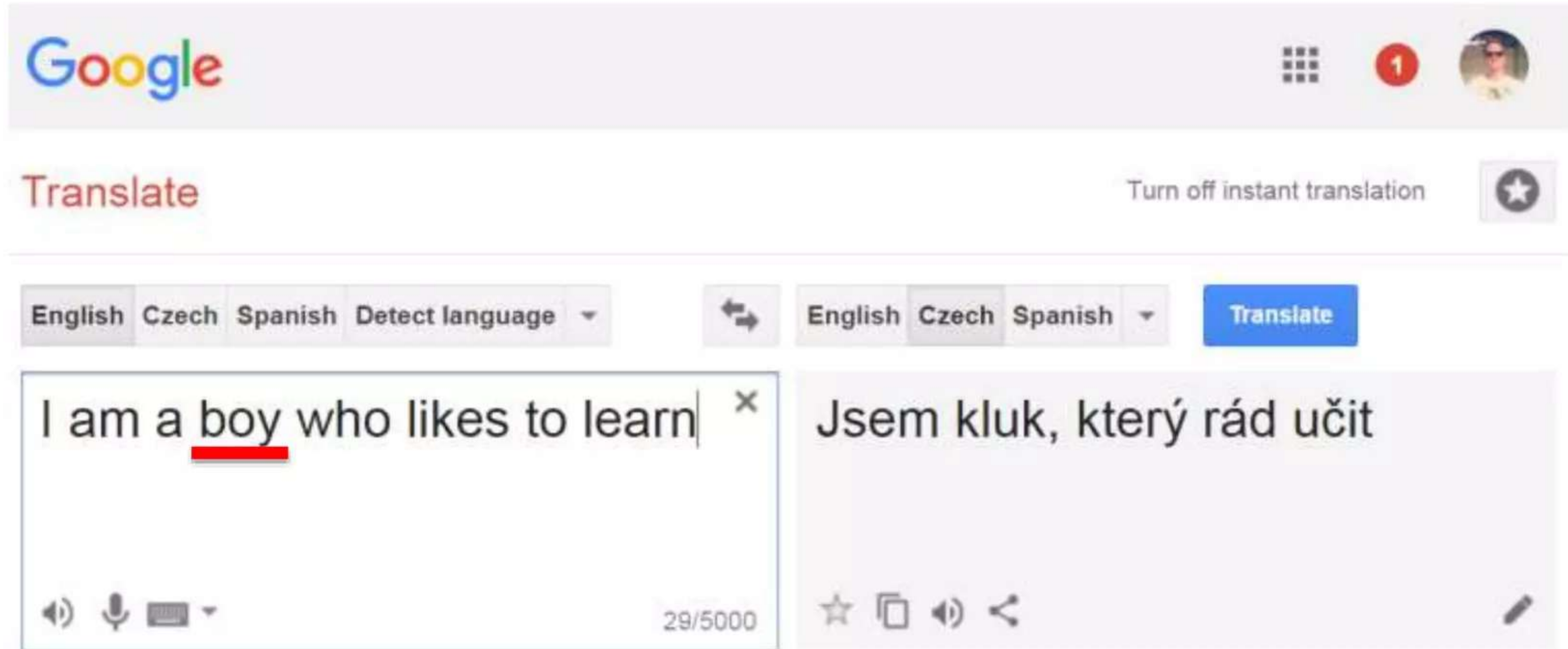
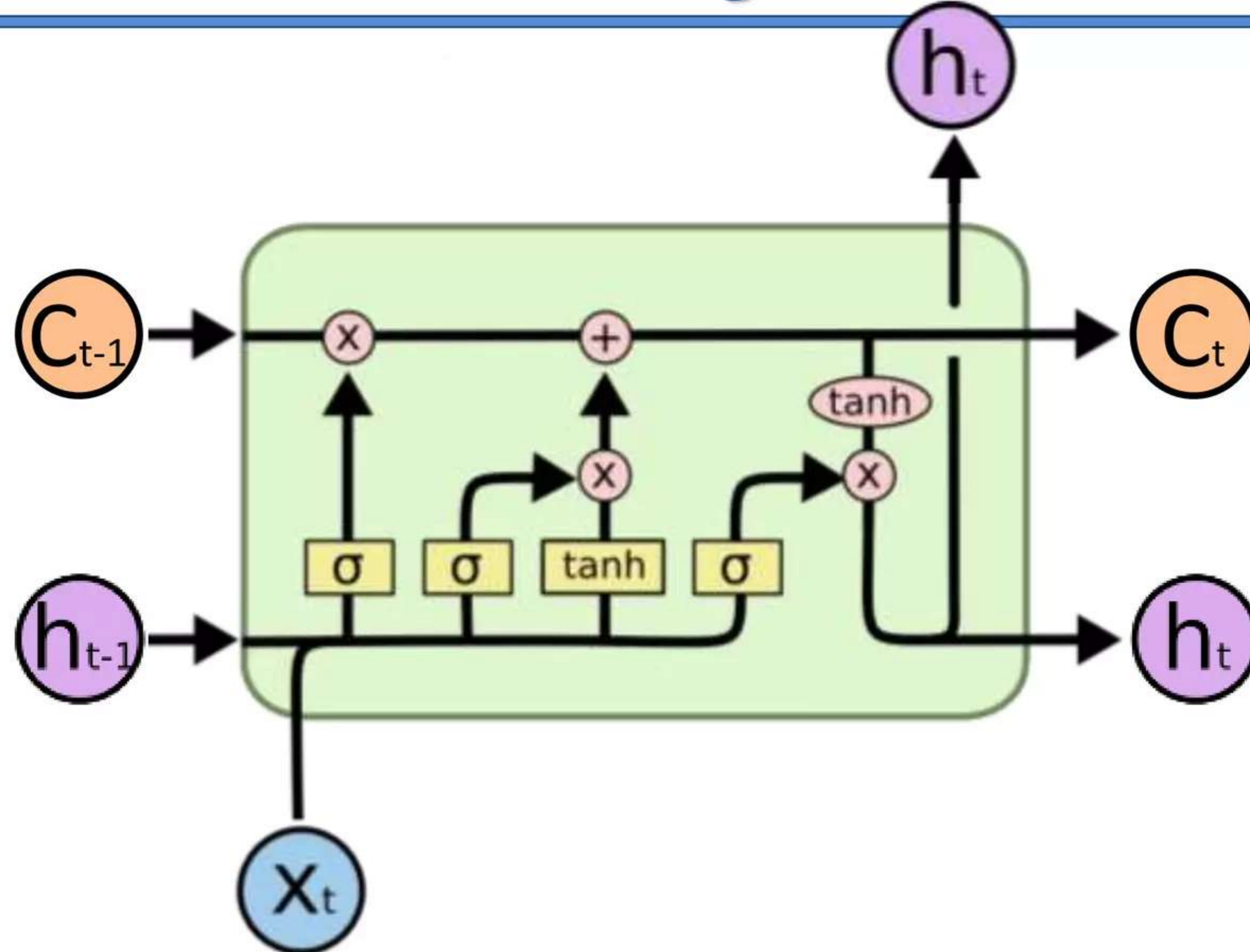


Image Source: colah.github.io

Long Short-Term Memory



Long Short-Term Memory



Long Short-Term Memory

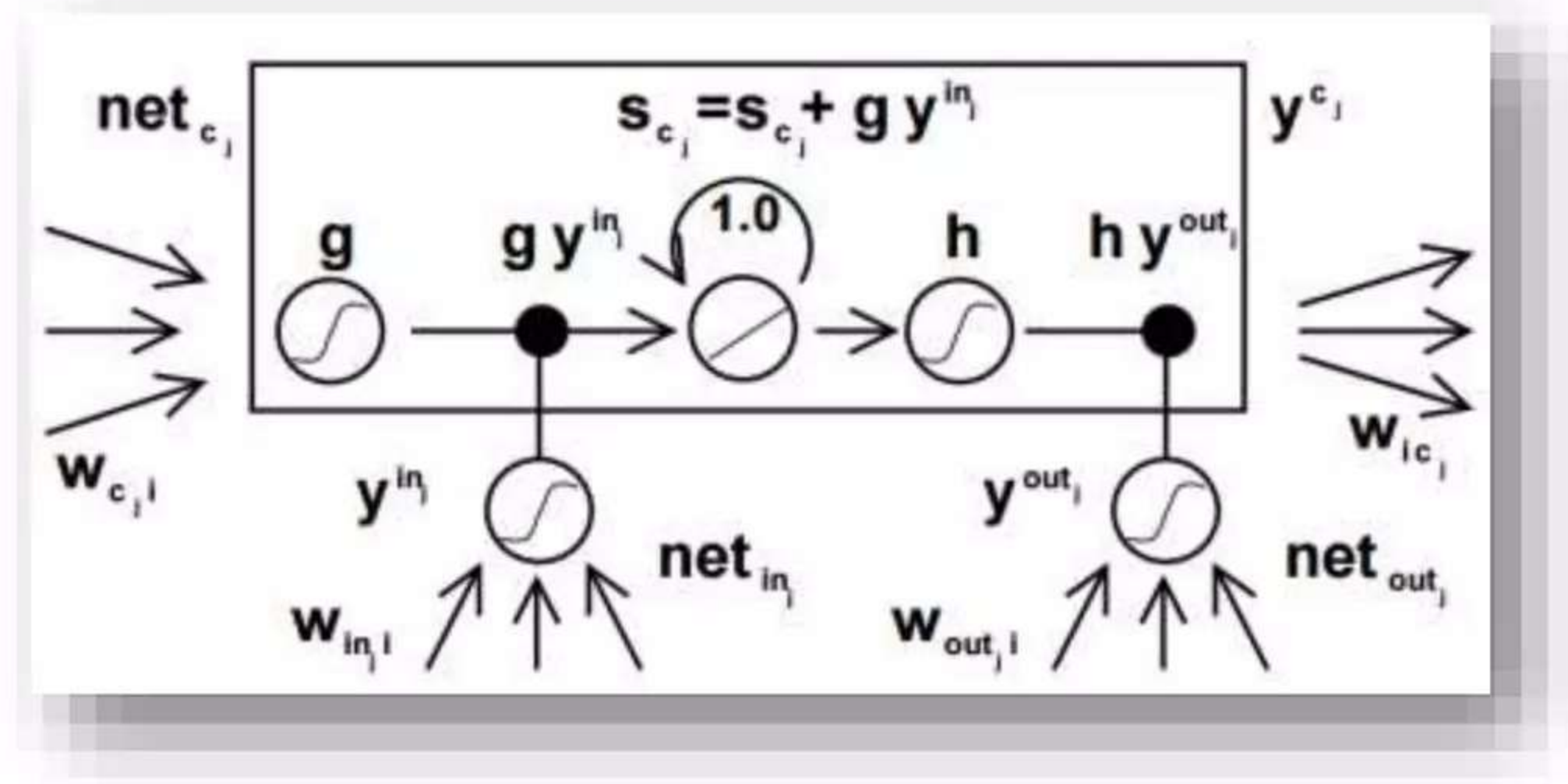
Additional Reading:

Long Short-Term Memory

By Sepp Hochreiter &
Jurgen Schmidhuber (1997)

Link:

<http://www.bioinf.jku.at/publications/older/2604.pdf>



Long Short-Term Memory

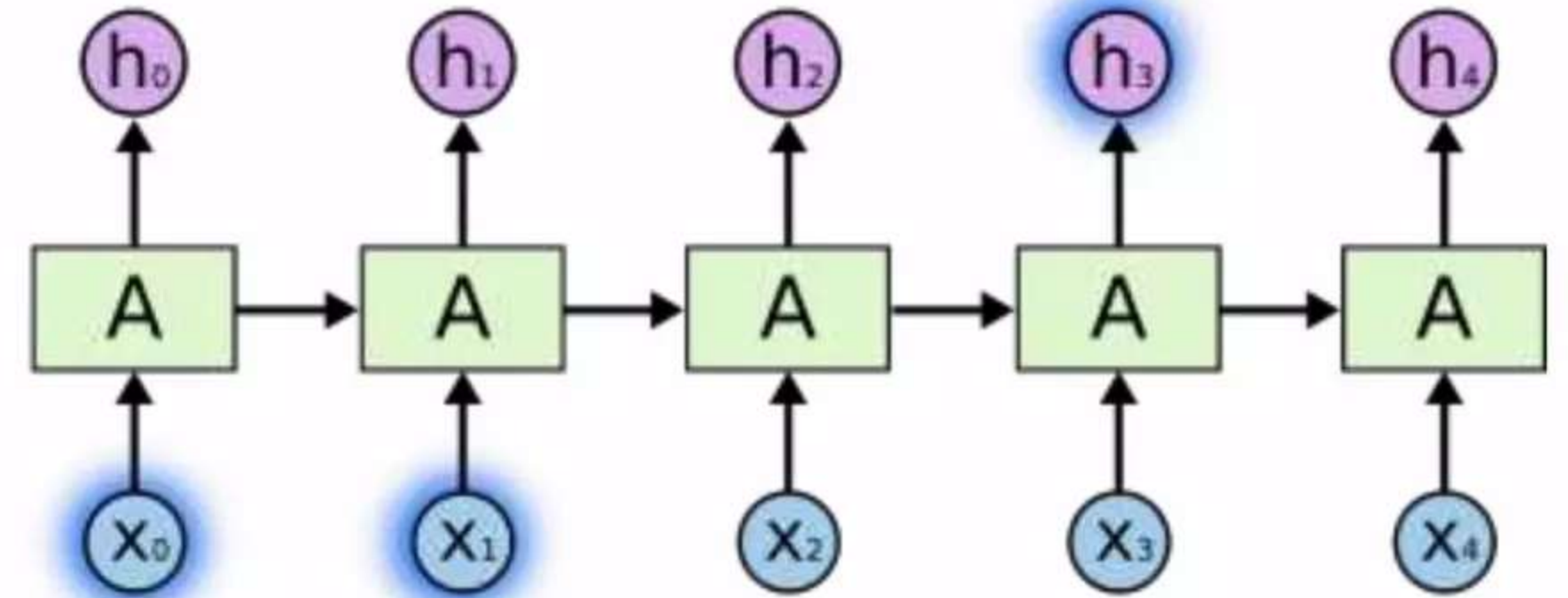
Additional Reading:

Understanding LSTM Networks

By Christopher Olah (2015)

Link:

<http://colah.github.io/posts/2015-08-Understanding-LSTMs/>



Long Short-Term Memory

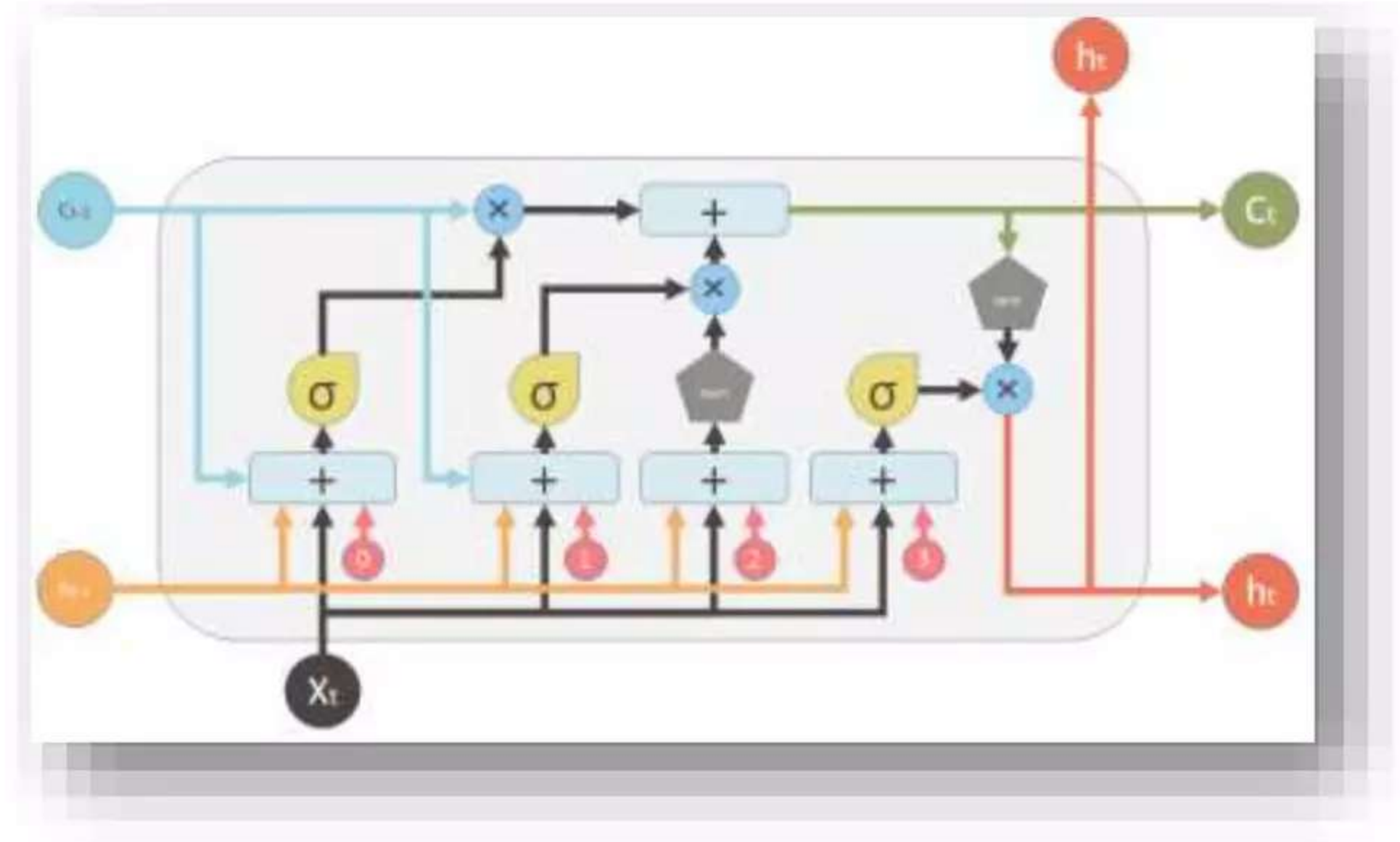
Additional Reading:

Understanding LSTM and its diagrams

By Shi Yan (2016)

Link:

<https://medium.com/@shiyang/understanding-lstm-and-its-diagrams-37e2f46f1714>



LSTM Practical Intuition

Long Short-Term Memory

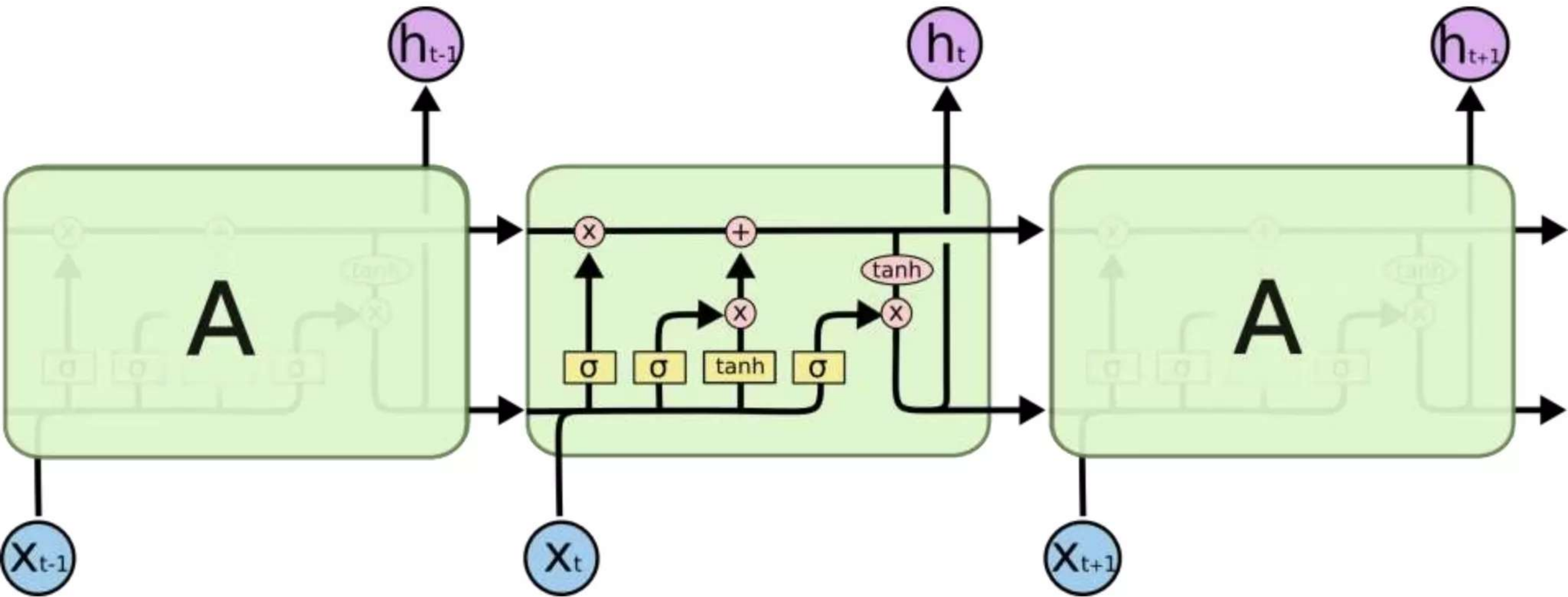


Image Source: colah.github.io

Long Short-Term Memory

Cell sensitive to position in line:

The sole importance of the crossing of the Berezina lies in the fact that it plainly and indubitably proved the fallacy of all the plans for cutting off the enemy's retreat and the soundness of the only possible line of action--the one Kutuzov and the general mass of the army demanded--namely, simply to follow the enemy up. The French crowd fled at a continually increasing speed and all its energy was directed to reaching its goal. It fled like a wounded animal and it was impossible to block its path. This was shown not so much by the arrangements it made for crossing as by what took place at the bridges. When the bridges broke down, unarmed soldiers, people from Moscow and women with children who were with the French transport, all--carried on by vis inertiae--pressed forward into boats and into the ice-covered water and did not, surrender.

Cell that turns on inside quotes:

"You mean to imply that I have nothing to eat out of.... On the contrary, I can supply you with everything even if you want to give dinner parties," warmly replied Chichagov, who tried by every word he spoke to prove his own rectitude and therefore imagined Kutuzov to be animated by the same desire.

Kutuzov, shrugging his shoulders, replied with his subtle penetrating smile: "I meant merely to say what I said."

Image Source: karpathy.github.io

Long Short-Term Memory

Cell that robustly activates inside if statements:

```
static int __dequeue_signal(struct sigpending *pending, sigset_t *mask,
                           siginfo_t *info)
{
    int sig = next_signal(pending, mask);
    if (sig) {
        if (current->notifier) {
            if (sigismember(current->notifier_mask, sig)) {
                if (!current->notifier(current->notifier_data)) {
                    clear_thread_flag(TIF_SIGPENDING);
                    return 0;
                }
            }
        }
        collect_signal(sig, pending, info);
    }
    return sig;
}
```

Cell that is sensitive to the depth of an expression:

```
#ifdef CONFIG_AUDITSYSCALL
static inline int audit_match_class_bits(int class, u32 *mask)
{
    int i;
    if (classes[class]) {
        for (i = 0; i < AUDIT_BITMASK_SIZE; i++)
            if (mask[i] & classes[class][i])
                return 0;
    }
    return 1;
}
```

Image Source: [karpathy.github.io](https://github.com/karpathy)

Long Short-Term Memory

A large portion of cells are not easily interpretable. Here is a typical example:

```
/* Unpack a filter field's string representation from user-space
 * buffer. */
char *audit_unpack_string(void **bufp, size_t *remain, size_t len)
{
    char *str;
    if (!*bufp || (len == 0) || (len > *remain))
        return ERR_PTR(-EINVAL);
    /* Of the currently implemented string fields, PATH_MAX
     * defines the longest valid length.
     */
}
```

Image Source: karpathy.github.io

Long Short-Term Memory

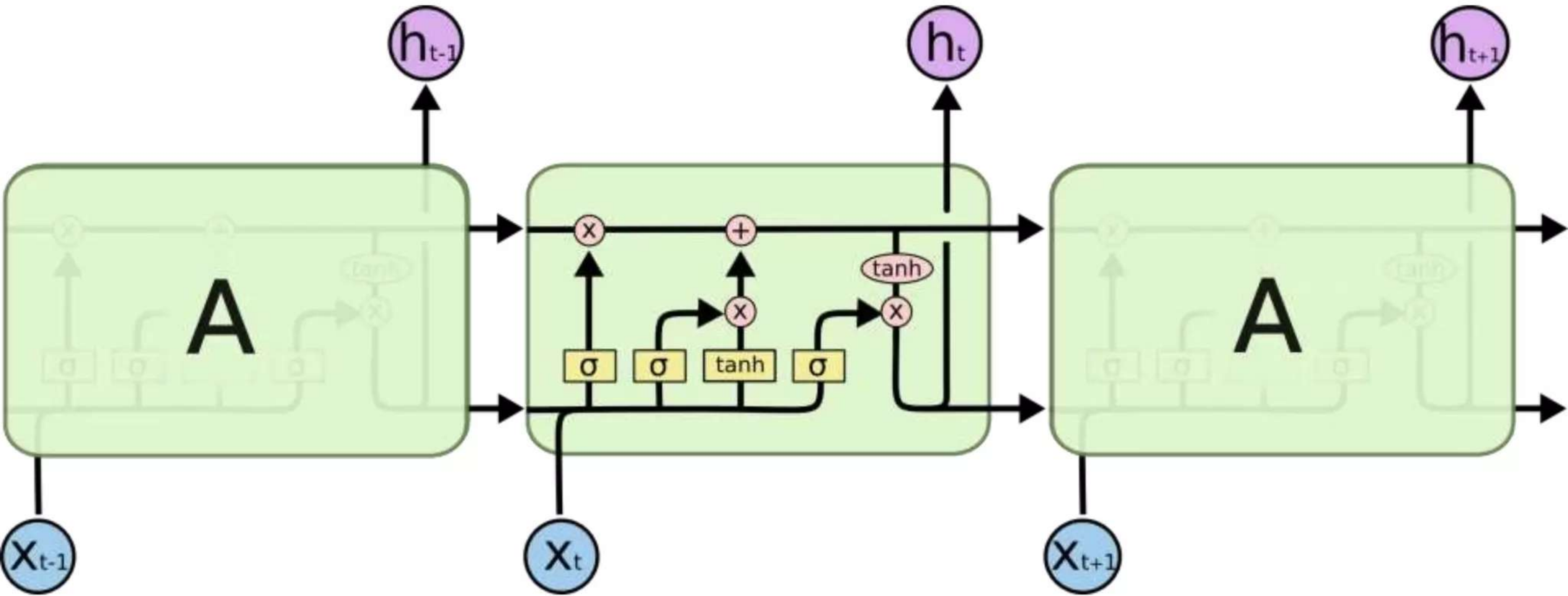


Image Source: colah.github.io

http://www.ynetnews.com/] English-language website of Israel
http://www.bacahets.com/ - xglis hlinguagesa irsite of ts lae
d : xne. waea. .awat o a. s & ntiaca - sard eelh oan tbi sanfanrei f
mw - 2 p i i i s o e s s i s . / e r n . c] (d c e e n e p e s a a i k i i e e l e d h , i r t h r a o n s
dr . < : a h b - n p t w t . x i g h / m a) T v d r y z i c o u e d l s u : t h a - o o t u , s t u i f
s t p , t c o a 2 d r u l w o c l e n s r] p . l l v a o d , , e y t c - n d m - o i b u v s] b b i m s u l t

gest newspaper ' ' [[Y e d i o t h A h r o n o t h]] ' ' ' Hebrew-language
e l a a w s p a p e r s o [[T e l t i (f e a n e m t i) ' ' * ' ' [e r r e w s l e n g u a g e :
i r s c o e e n a i T T h A o a i n n h S r m u w] e y s [' i n e i a ' s i w d d e ' h s
u s . . s e t l g o r s . a s a t C a r e e g ' a C l r i s z] i e ' : : , # : T A a a a a t B a s e e i l o '
- t u a e v r t i d , t B A m S u s y u t]] A s a o i g s]] , . : s M B o l o u s : T o u a - n : d
a , d , i i u i t i c p .] (l S v H v t u s u i e D n o e g a n o . ,] : { C C u i b o h e C y b k s l s : r

icals : ' ' ' * ' ' [[G l o b e s]] ' ' [http://www.globes.co.il/] busi
cal : ' ' ' * ' ' [T a a b a] ' ' ([t t p : / / w w w . b u o b a l . c o m u n / s A - y t i r
s t l ' [h A e o v e l t s a h a d : x g e . w a o i r . r t o a . e l . i T & a i e
t t ' ' ' & [& & m C o e r o n e ' : : , i ' o d w . , : n i i i s a a u e . e n i / o m l c C . (e f t g
a ' n : , C : & : # * : a f D r u s u] l , . o m e l p < , d h a ; d e u o o t / i h n c s i f S , u r h c
n k i <] : & 1 1 s T G u i t r s i , : b a c m r - x t p o b - g r e s i s l e r l n a f a D] l o s p t

ily * ' ' [[H a a r e t z | H a ' A r e t z]] ' ' [http://www.haaretz.co.il/]
ly * [[T e r r d n F e r a n t a h]] ' ' ([t t p : / / w w w . b o n m d s t . c o m u n / s -
r e ' ' h A i l n n t t e H a l s r c n o l ' s a h a d : x n e . w a a m r t d h e o h . o l . c &
k i . : * s C O S a n l t h i T i m ' l i] e : , i m c d w - 2 p h i
ds - ! [t B T C o m m g d]] W o n a a e , : . b a e r r . < t a i b - d u l c n n c / a r n e s i] l
nds # & : G l D u v c c s a o S u c l t e l l z l . : o ' o m t l . : e o a 2 n i v f s r o o e i u n a l a)

Image Source: karpathy.github.io

Additional Reading

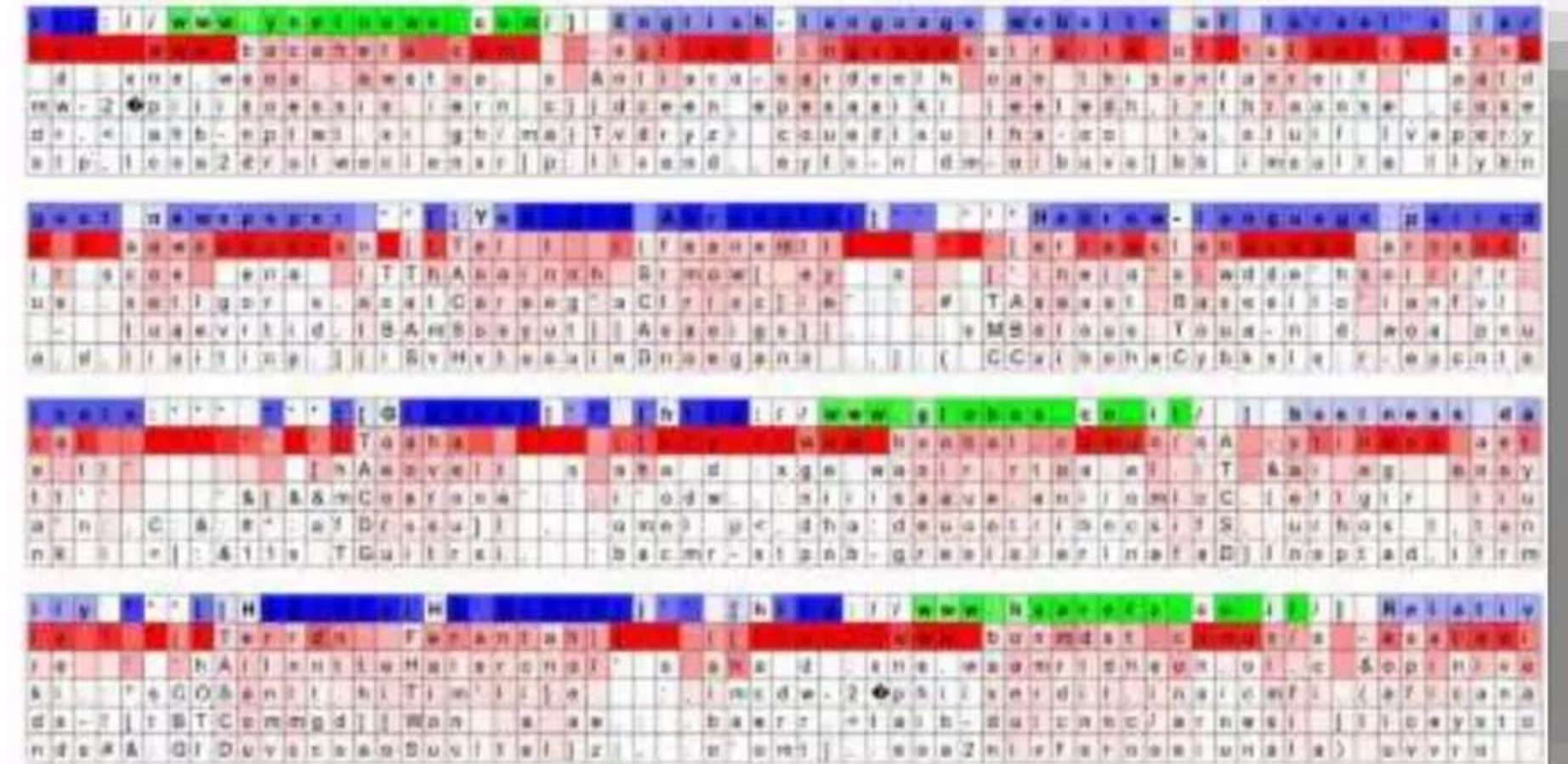
Additional Reading:

The Unreasonable Effectiveness of Recurrent Neural Networks

By Andrej Karpathy (2015)

Link:

<http://karpathy.github.io/2015/05/21/rnn-effectiveness/>



Additional Reading

Additional Reading:

Visualizing and Understanding Recurrent Networks

By Andrej Karpathy et al. (2015)

Link:

<https://arxiv.org/pdf/1506.02078.pdf>



LSTM Variations

LSTM Variations

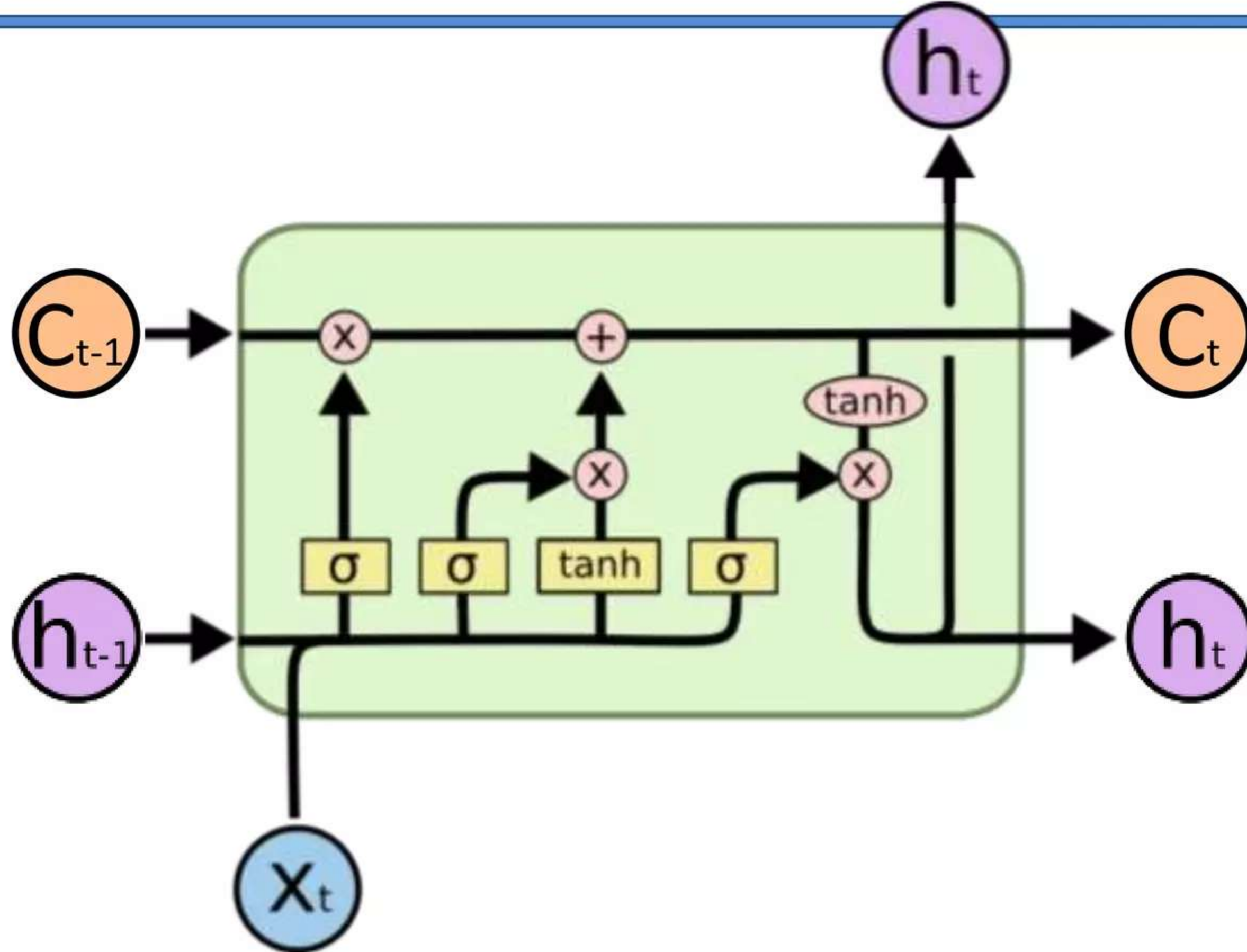


Image Source: colah.github.io

LSTM Variations

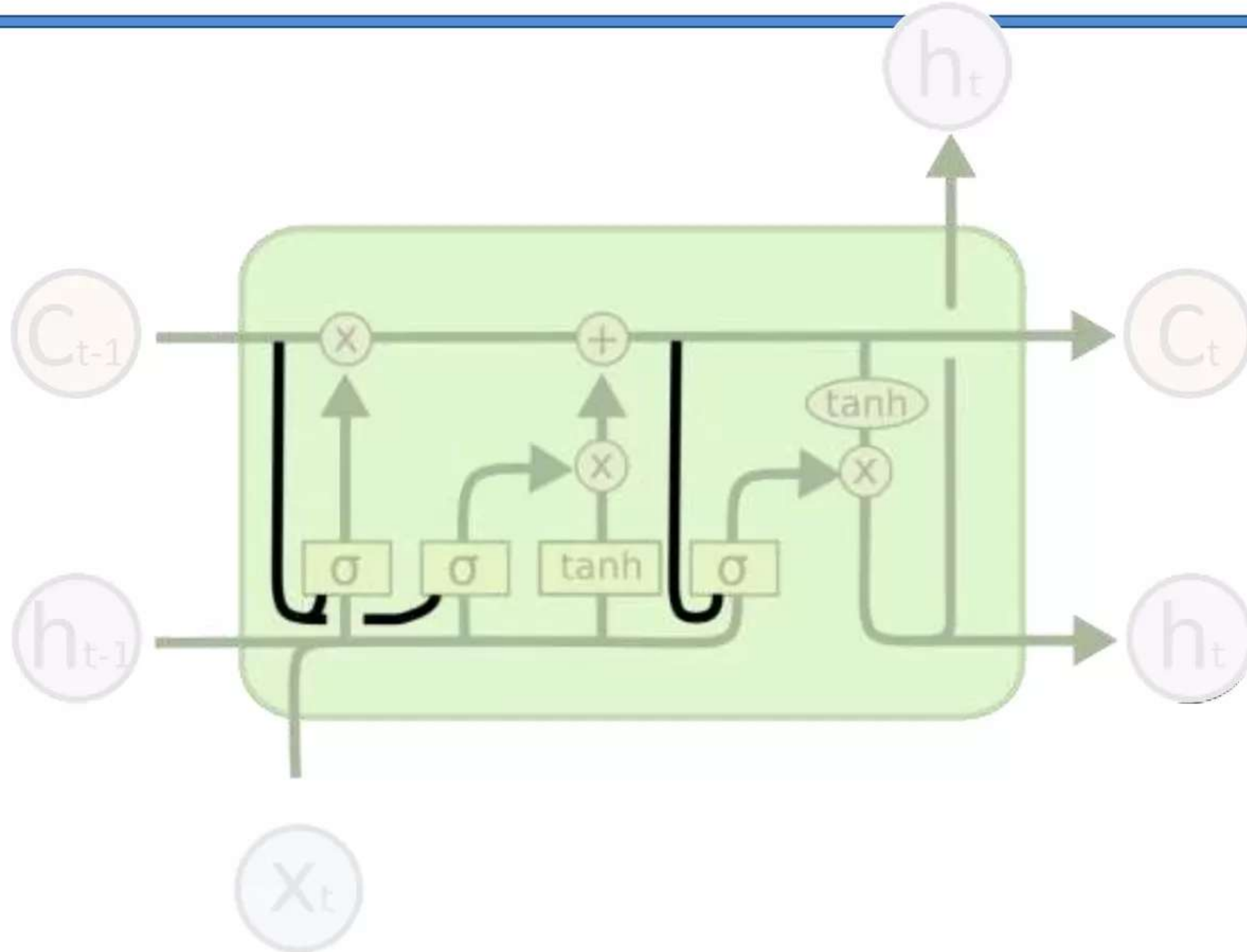


Image Source: colah.github.io

LSTM Variations

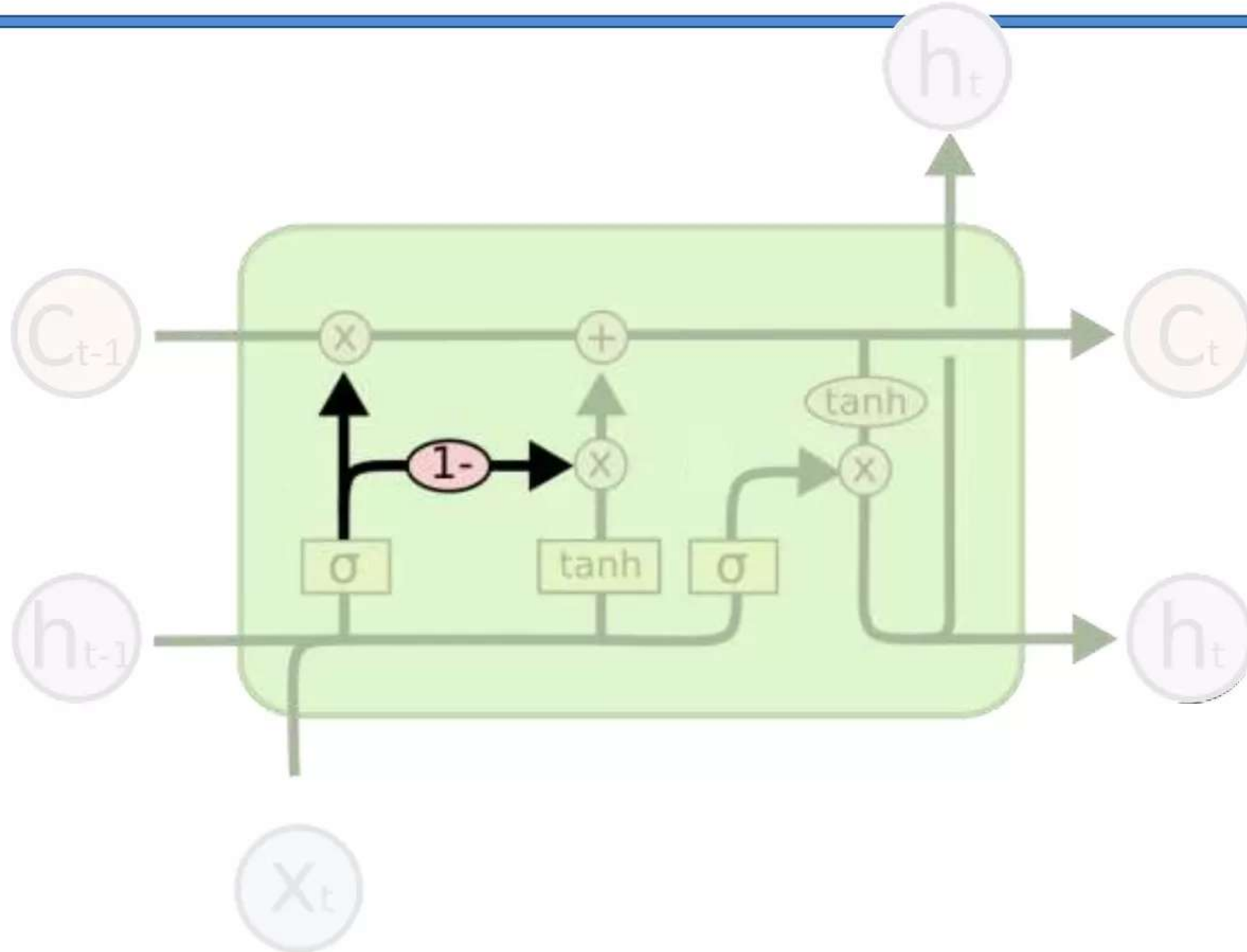
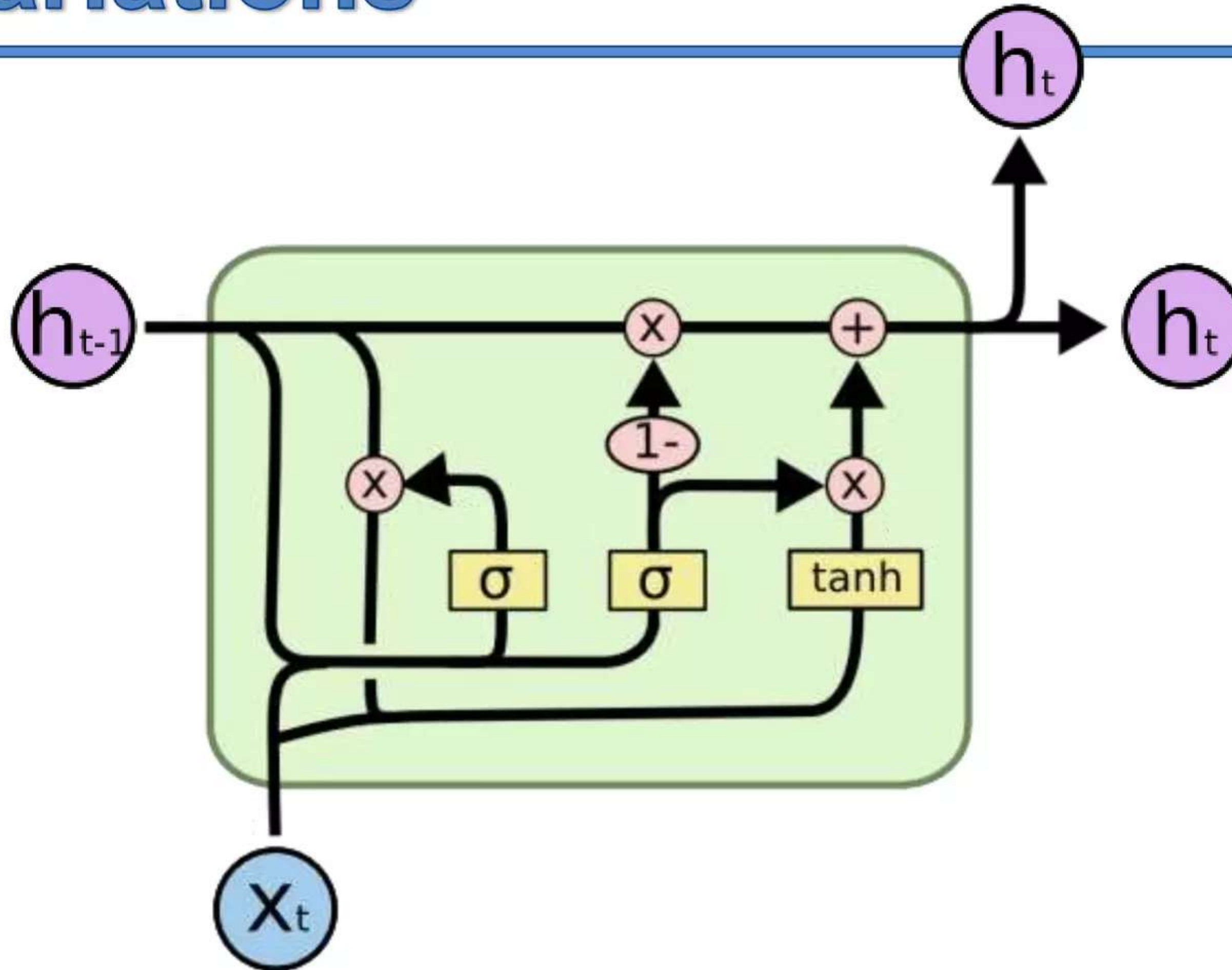


Image Source: colah.github.io

LSTM Variations



Long Short-Term Memory

Additional Reading:

LSTM: A Search Space Odyssey

By Klaus Greff et al. (2015)

Link:

<https://arxiv.org/pdf/1503.04069.pdf>

